Python for Data Analysis

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1 Python for Data Analysis

These are the materials which are used for a course on *Python for Data Analysis* at the University of Freiburg in summer 2026. It does not assume any prior knowledge of python (although this might help). The goal of the course is some literacy in python, including basic knowledge such as data types and control flow, but also using libraries and APIs, reading and dealing with data, up to some methods such as PCA and regression.

Basically, the course has three parts: 1. Python basics: data types, functions, loops, conditions etc; (Lessons 01, 02, 03, 04) 2. Tools for data analysis: numpy, matplotlib, pandas; (Lessons 05, 05, 07, 08) 3: Project: choose data that interests you and creat a dashboard

Within the first two parts, you will be assigned homeworks, which you can find at the end of each section. For the third part, you will have to find a project by yourself.

Let us discuss some preliminairies.

Your python environment: We will assume that python and jupyter is installed on your system. These course notes are written with python 3.13, but other versions might work as well. In addition to plain `python`, we will need some libraries. Which ones we need (including their versions) is written in the file `requirements.txt`. Some jupyter notebooks might not work if these are not installed on your system. (To be more precise, they must be installed on the jupyter server which runs the notebools.)

General note: We avoid repeating the distinction between the commands `python` and `python3`, and will simply write `python3` in the sequel.

Starting a Jupyter notebook:

There are basically three ways `python` commands can be run: 1. In interactive mode on the command line by typing `python3` in your shell, 2. In `python`-scripts, usually ending with `myScript.py`. You can run these using `python3 myScript.py`. 3. Jupyter notebooks have the opportunity to mix usual text with `python` commands. We will use these in this course.

For Jupyter notebooks, there are at least two ways to run them:

1. Start the Jupyter server using `jupyter lab` (hopefully, it is already installed), which opens your browser where you can edit and save what you have done;
2. Use `vscode` as you IDE and work from there; see below.

Integrated Development Environment (IDE): When programming, we need to type code. This is possible in a code editor, and best done in a more sophisticated environment such as `vscode` or pycharm. It is best to acticvate your vertual environment using `source venv/bin/activate` before starting your IDE. In order to start vscode, go to the main folder of the course notes, and type `code .` E.g., vscode can be used for all three ways to write Python commands:

1. Use `Terminal -> New Terminal` in order to open a terminal, type `python3` and you are in interactive mode,
2. Write a Python script and run it using the opoened terminal,
3. Click on some Jupyther notebook, and vscode starts the Jupyter server for you.

A first Python cell: Here is the first cell of Python-code:

```
x = "Hello, "
y = "world!"
print(x+y)
```

```
Hello, world!
```

If you are working within the jupyter notebook (and not just in the html-file), you can execute this piece of code: Press the play bottom on the left (or hit `Control + Enter` within the Python-cell).

Using AI (ChatGPT, Gemini etc, Copilot, Claude, etc):

During the last years, AI has increasingly improved in code writing. You are encouraged to use this new technology! As usual when learning new things, this new tool can help a lot, and will be able to solve the exercises within the Jupyter notebooks. However, you still have to undestand the code, and what might go wrong during execution. As a first course in Python, you will end up having up to a few hundred lines of code, which has to be coherent. You must know the structure of that code, and in order to extend it, fix bugs etc. If you are ever involved in bigger projects, this is even more important.

For these course notes, I have also used AI (chatGPT, gemini) in order to ask the AI which topics I should cover, and make concrete suggestions for a workflow. (While I have been programming in Python3 for some years now, my experience with `numpy` and `matplotlib` was still limited when I started to write the course, since I have used `R` for such topics in the past.) Not all suggestions from the AI were great. However, I think that the end result for that course is much better than without the help of AI. I hope this also helps the learning experience. In particular, an earlier version of Chapter 11 (generated with ChatGPT) served as a source of material: its content on `numpy`, `pandas`, and `scipy.stats` was integrated into Chapters 6, 7, and 8 with the help of Claude Code, while the machine learning portions remain in the archived original.

2 Simple Data types

Download notebook.

2.1 Overview

Every variable or other object in python has a type. This is the kind of information which is stored insice the variable or object Here is an overview over the most important simple data types.

Type	Example	Description
`int`	`5`	Integer
`float`	`3.14`	Floating-point number
`bool`	`True`	Boolean value
`str`	`"hello"`	Text string
`NoneType`	`None`	No value / null value

- You can find about the type of any expression `x` in Python using `type(x)`.
- In fact, `type(x)` is already the first example of a built-in-function. The function `type` takes a single value as input, and returns its type. We will encounter various more functions already in this section. We will learn more about functions (built-in and self-written) in Section 3.4.
- There are conversions between some types. For example, `"0"` is a string, but `int("0")` (again a function, taking a single value, and returning its integer value) the integer `0`. Conversely, `str(0)` is the string `"0"`. We will have more examples below.

2.2 Integers (`int`)

With `int`s (numbers …,-2,-1,0,1,2,…), you can compute as usual.

```
x = 7
y = 2
print("x + y = ", x + y)
print("x * y =  ", x * y)
```

```
x + y =  9
x * y =   14
```

```python
# x to the power of y
print("x ** y = ", x ** y)
```

```
x ** y =  49
```

```python
# x divided by y (which gives a `float`)
print("x / y = ", x / y)
```

```
x / y =  3.5
```

```python
# integer-valued division of x by y
print("x // y = ", x // y)
```

```
x // y =  3
```

```python
# x modulo y
print("x % y = ", x % y)
```

```
x % y =  1
```

In most programming languages (e.g. `C++`), there is a minimal and a maximal possible value for an int. Python, however, does not have this limitation. Only the amount of available memory restricts the numbers which can be stored in an `int`.

2.3 Floating point numbers (`float`)

Generally speaking, floats are decimal numbers, which come with an exponent and a mantissa, e.g. $1.2 \cdot 10^2$. Python can do the same computations as with ints, but the following should be kept in mind:

- If you type `1`, Python thinks this is an int, but if you type `1.0` or `1.`, Python thinks this is a float.
- Unlike ints, floats do have a maximal and minimal element. The reason is that every float is stored in 64 bits, one for the sign, 11 for the exponent and 52 for the mantissa. As a result, floats range from about 10^{-324} to 10^{308}.

- When doing mixed computations with floats and int, e.g. `1.0 + 1`, the int is changed into a float.
- It is not possible to perform exact computations with some floats; see the example below.

```
x = 1.0
y = 1
print("x + y = ", x + y)
```

```
x + y =  2.0
```

```
print("0.1 + 0.2 = ", 0.1 + 0.2)
```

```
0.1 + 0.2 =  0.30000000000000004
```

Special floats are `inf`, `-inf` and `nan` (not a number). Let us see them in action:

```
print((10. ** 308) * 10)
```

```
inf
```

```
print((10 ** 308) * 10)
```

```
100000000000000
```

```
print(int(0.0))
```

```
0
```

```
print(int(0.5))
```

```
0
```

If you deliberately want to introduce `nan` or `inf`, you can do this:

```
print(float("inf"))
print(float("nan"))
```

```
inf
nan
```

On `float`, you can use many functions. We only name two at the moment, for some float `x`:

- `abs(x)`: The absolute value of `x`.
- `round(x)` and `round(x,n)`: Round `x` to the next `int`, or after `n` digits.

In Section 5.2, we will encounter that many more usual functions (like `sqrt`) can be used on floats, but we need to `import` the mathematical library for this.

2.4 True-False variables (`bool`)

There are exactly two boolean values, known as `True` and `False`. You can compute with them using `not`, `and`, `or` and 'xor'. In addition, `True` and `False` can be the result of a condition, such as the expression `1 == 2`.

```
x = True
y = False
print("not x =", not x)
```

```
not x = False
```

```python
print("x and y =", x and y)
```

```
x and y = False
```

```python
print("x or y =", x or y)
```

```
x or y = True
```

```python
print("x xor y =", x ^ y)
```

```
x xor y = True
```

```python
print("((not x) == y) =", (not x) == y)
```

```
((not x) == y) = True
```

```python
print("(1 == 2) =", 1 == 2)
```

```
(1 == 2) = False
```

```python
# Python converts `bool` into `int`
print("x + y =", x + y)
```

```
x + y = 1
```

```python
# != means "not equal"
print("(1 != 2) =", 1 != 2)
```

```
(1 != 2) = True
```

```python
# < refers to less than
print("(1 < 1) =", 1 < 1)
```

```
(1 < 1) = False
```

```
# <= refers to less than or equal
print("(1 <= 1) =", 1 <= 1)
```

```
(1 <= 1) = True
```

```
z = float("nan")
print("(z == z) =", z == z)
```

```
(z == z) = False
```

2.5 Text (`string`)

In many cases, strings are objects in quotation markes, such as in "'Hello, world!".

- **Operations on strings**: You can concencatenate strings using `+` as in "'Python" + "for" + "Data" + "Analysis" (but you cannot subtract strings). You can also repeat a string by multiplying it with an int.
- **f-strings**: They were introduced in Python 3.6. They are the way to go if you want to mix description and variables. They are written as `f"Hello, {name}"`. The `f` introduces the f-string. Within curly braces, there is a variable, which should be printed instead of the variable name.
- **Special characters**: There are many of them, we only mention `\t` (tabulator) and `\n` (new line) here.
- **Multiline strings**: Usually, python interprets a new line such that all commands are finished. However, if you want to use a sttring spanning multiple lines, you don't want this. Multiline strings start and end with `"""` (a triple quotation mark).
- **Quotation marks**: In various cases, you can use `'` instead of `"`. Sometimes this is useful when nesting quotation marks.
- **Substrings**: We will learn more about substrings, but here is already a way to see if a substring occurs within a string or not. The value of `"th" in "Python"` amount to `True`.

```
x = "Hello,"
y = "world"
# Concatenating two (or more) strings
print(x + y + "!")
```

```
Hello,world!
```

```python
# repeating a string
z = "bla"
print(z*3)
```

```
blablabla
```

```python
# a special character
print("\U0001F60E")
```

```python
# an example of an f-string
name = "Alice"
print(f"Hello {name}  ")
```

```
Hello Alice
```

```python
# a multiline string
text = """This is a multiline string.
It can span several lines.
Each line break is preserved."""
print(text)
```

```
This is a multiline string.
It can span several lines.
Each line break is preserved.
```

```python
# determine if a given substring occurs within a string
print("Is 'th' a substring of 'Python'? ", 'th' in 'Python')
print("Is 'the' a substring of 'Python'? ", 'the' in 'Python')
```

```
Is 'th' a substring of 'Python'?  True
Is 'the' a substring of 'Python'?  False
```

For strings, there are various build-in functions. We mention some of them. Here, `s` is a string.

- `str.strip(s)`: Deletes spaces in the front and back.

- `len(s)`: gives the length of the string.
- `str.replace(s, "x", "y")`: Replaces each occurrence of `"x"` in `s` by `"y"`.
- `str.split(s, "x")`: Splits `s` at each occurrence of `"x". The result is a` list‘, which we will encounter in Section 4.1. “,”.join
- `str.lower(s)`: Return the same string, but all letters are lowercase.
- `str.upper(s)`: Return the same string, but all letters are uppercase.
- `str.isalpha(s)`:
- `str.isdigit(s)`: Returns `True` if the string is actually a number, and `False` otherwise.
- `str.isalnum(s)`: Returns `True` if the string consists only of letters and numbers, and `False` otherwise.
- `str.isspace(s)`: Returns `True` if the string consists only empty spaces, and `False` otherwise.
- `str.islower(s)`: Returns `True` if the string consists only of lower case letters, and `False` otherwise.
- `str.isupper(s)`: Returns `True` if the string consists only of upper case letters, and `False` otherwise.
- `str.count(s, "x")`: Counts how ofter `"x"` appears in `s`.

Here is another very useful notation convention, known as dot-notation: For a function `str.something(s,...)`, where `s` is of type `str`, you can as well write `s.something()`. For example, `s.strip()` is the same as `str.strip(s)`.

2.5.1 Formatting numbers in f-strings

claude generated

When printing numerical results, we often want to control the number of decimal places, alignment, or padding. Inside an f-string, a format specifier follows the variable after a colon: `f"{x:.4f}"`.

- `f"{x:.4f}"`: float with 4 decimal places.
- `f"{x:.2e}"`: scientific notation with 2 decimal places.
- `f"{x:>10}"`: right-align in a field of width 10.
- `f"{x:<10}"`: left-align in a field of width 10.
- `f"{n:,}"`: integer with thousands separator.
- `f"{x:.2%}"`: format as percentage with 2 decimal places.

```
import math
pi = math.pi
print(f"pi = {pi}")
print(f"pi = {pi:.2f}")
print(f"pi = {pi:.8f}")
print(f"pi = {pi:.2e}")
```

```
pi = 3.141592653589793
pi = 3.14
pi = 3.14159265
pi = 3.14e+00
```

```
# alignment and padding
for name, score in [("Alice", 92.5), ("Bob", 87.123), ("Charlie", 100.0)]:
    print(f"{name:<10} {score:>8.2f}")
```

```
Alice         92.50
Bob           87.12
Charlie      100.00
```

```
# thousands separator and percentage
n = 1234567
ratio = 0.8734
print(f"n = {n:,}")
print(f"ratio = {ratio:.1%}")
```

```
n = 1,234,567
ratio = 87.3%
```

This is especially useful when printing tables of results in data analysis.

2.6 Missing things (`NoneType`)

When analysing data, it is common to have missing values. These usually come as `None`. This is different from anything else we encountered so far (including `nan` and `False`).

```
x = None
print(x)
```

```
None
```

```
print("(x is None) =", x is None)
```

```
(x is None) = True
```

```python
print("(None == False) =", None == False)
```

```
(None == False) = False
```

```python
print("(None == nan) =", None == float("nan"))
```

```
(None == nan) = False
```

2.7 Exercises

Exercise 1: Find out how to use scientific notation in Python. (If you don't know what this is, please find out!)

Exercise 2: Is there a Python package which can compute `0.1 + 0.2 == 0.3` without error?

Exercise 3: Print `"Hello, world!"` including the quotation marks in the output.

Exercise 4: Do newlines and tabs `\n`, `\t` count in `str.isspacce()`? How do numbers interact with `str.islower()`? If `s` is the strign of a number, what is `s.lower()`? Play around a bit with these fucntions and edge cases you might see in data.

Exercise 5: If you are on Windows, it is annoying that file paths come with `\`, which is interpreted as the start of a special character. Find out two ways how to give the string of the file path `C:\Users\Alice\Documents\file.txt`.

Exercise 6: What is `bool(None)`? What is `int(None)`?

Exercise 7: In tables, we often have `""` for a missing value. Is this the same as `None`?

3 Control flow – conditions, loops, functions, etc

Download notebook.

Coding starts to become interesting when we can control how variables change, and repeat operations.

3.1 Conditions (`if ... elif ... else`)

As a simple (mathematical) example, let us compute the minimum of two numbers., which works by an `if`-condition:

```
a = 2
b = 5
if a < b:
    res = a
else:
    res = b
print(f"The minimum of {a} and {b} is:", res)
```

```
The minimum of 2 and 5 is: 2
```

In general, the structure of an `if`-statement reads:

```
if _condition1_:
    case1
elif _condition2_:
    case2
elif _condition3_:
    case3
...
else:
    case_else
```

Here, read `elif` also `else if`, which means that the code jumps into this case if the respective condition is the first to hold. The `else` case only applied if both the `if` and all `elif` conditions do not apply.

In fact, there are abbreviations in python3 which are very handy. In particular, look at this piece of code.

```
res = a if a < b else b
print(f"The minimum of {a} and {b} is:", res)
```

```
The minimum of 2 and 5 is: 2
```

Let us make another example, also using `elif`. Given some `n`, the task is to print `Fizz` if `n` is a multiple of 3, `Buzz`, if it is a multiple of 5, and `FizzBuzz` if both conditions are satisfied.

```
n = 165
res = ""
if n % 3 == 0 and n % 5 == 0:
    res = "FizzBuzz"
elif n % 3 == 0:
    res = "Fizz"
elif n % 5 == 0:
    res = "Buzz"
print(res)
```

```
FizzBuzz
```

3.2 Loops with fixed length (`for`)

There are `for`-loops in Python3, as discussed in this section, and `while`-loops in the next section. The basic use-case of a `for`-loop is repeating a task several times. As a simple example, adding `b` a number of `a` times, results in `a * b`.

```
a = 3
b = 7
res = 0
for i in range(a):
    res = res + b
print(f"{a} * {b} = {res}")
```

```
3 * 7 = 21
```

Importantly, `range(a)` is a synonym for the numbers `0,..., a-1`. (In particular, note that Python3 usually starts counting at `0` and not at `1`, which will also be important in the next chapter.)

The general structure of a `for`-loop is as follows:

```
for _var_ in _iterable:
    _sequence of instructions_
```

3.2.1 Greatest common divisor I

Let us look at a more interesting example, computing the greatest common divisor (gcd) of two numbers, `a` and `b`. A very simple algorithm is going through all numbers `1,...,` in order, and update result each time we encounter a number which divides both, `a` and `b`. (Here, `range(1,a+1)` is a synoym for the numbers `1,...,a`.)

```python
# Basic algorithm for finding the gcd of two numbers
a = 105
b = 33

res = 1
for i in range(1,a+1):
    if a % i == 0 and b % i == 0:
        res = i
print(f"The gcd of {a} and {b} is {res}.")
```

```
The gcd of 105 and 33 is 3.
```

Let us make some remarks on `for`-loops: * In the last example, note that the instructios within the `for`-loop depend on `i`. * Within a `for`-loop, you can use `continue` in order to stop the execution of the current iteration, and immediately jump to the next `i`. * Within a `for`-loop, you can use `break` in order to stop the whole execution of the `for`-loop.

3.3 Loops with unknown length (`while`)

The `for`-loop directly iterates over some variable (`i` above). The `while`-loop instead comes with a condition, which is checked everytime it is started new. The general structure is:

```
while _condition_:
    instructions which might change the value of _condition_
```

Let us make two examples: finding $\sqrt{2}$ and computing the gcd using Euclid's algorithm.

3.3.1 Finding $\sqrt{2}$

For finding $\sqrt{2}$, note that this is a fixed point for the iteration

$$x_{n+1} = \frac{x_n}{2} - \frac{1}{x_n}.$$

(In order to see this, assume that $x = x_n = x_{n+1}$, and multiply the recursion by x.) We can use this as follows.

```python
x = 1
eps = 1e-10
x_prev = 0

while abs(x - x_prev) > eps:
    x_prev = x
    x = x/2 + 1 / x
print(x)
```

```
1.414213562373095
```

3.3.2 Finding the greatest common divisor II

The above algorithm (using `for`-loops) for finding the gcd were not particulary efficient. (You might want to test them with some large numbers.) Euclid's algorithm is much more efficient. It is based on the observation that `c % a == 0` (`c` divides `a`) and `c % b == 0` iff `c % a == 0` and `c % (b % a)`.

In order to see this, note that `c % a == 0` and `c % b == 0` iff `c % a == 0` and `c % (b - da) == 0` for any `d`. Then, the result follows from choosing `d = b \\ a` since `b - (b \\ a) * a = b % a`. Moreover, every number divides `0`, so if `b % a == 0`, the gcd of `a` and `b` is `a`.

We can use these insights, taking `a < b` for simplicity. Starting with `a` and `b`, we compute if `b % a == 0`. If yes, we are done, and return `a`. If not, we compute the gcd of `a` and `b % a` instead.

```
# Euclid's algorithm for finding the gcd of two numbers
a_input = 105
b_input = 33
# Copy the input to new variables
a = a_input
b = b_input
# Run Euclid's algorithm
while a != 0:
    r = b % a
    print(f"b = {b}, a = {a}, remainder = {r}")
    b = a
    a = r
print(f"The gcd of {a_input} and {b_input} is {b}.")
```

```
b = 33, a = 105, remainder = 33
b = 105, a = 33, remainder = 6
b = 33, a = 6, remainder = 3
b = 6, a = 3, remainder = 0
The gcd of 105 and 33 is 3.
```

3.4 Functions (`def`)

Once we implement something, we wan to reuse it without re-writing the code. This means, we want to pack our code into its own function for reusing it. For Euclid's algorithm, this would look as follows:

```
a = 105
b = 33

def gcd(a,b):
    """ Compute the greatest common divisor of a and b."""
    while a != 0:
        r = b % a
        b = a
        a = r
    return b
```

```python
print(f"The gcd of {a} and {b} is {gcd(a,b)}.")
```

```
The gcd of 105 and 33 is 3.
```

The general structure is

```
def name_of_the_funcion(arg1, arg2, arg3 = default3, arg4=default4):
    some instructions
    return something
```

Not all functions return something. Those who don't usually produce some output. Here is an example

```python
def greeting(name, prefix="Hello"):
    """ Return a greeting for name."""
    print(f"{prefix}, {name}!")

greeting("Alice")
greeting("Bob", prefix="Hi")
greeting(prefix="Welcome", name="Charlie")
```

```
Hello, Alice!
Hi, Bob!
Welcome, Charlie!
```

Here are some important remarks on functions:

- Variables within functions are private to them. (There will be exceptions to this) In the gcd example, variable names `a` and `b` occur both, within and outside of the function. Although the variables `a` and `b`, which are private to the function, are changed within the function, the `print`-statement still knowns their value from before the function definition.
- Variables which are given to functions might have default values, as the `prefix = "Hello"` in the `greeting` example.
- For giving variables, there are two options:
  - you can rely on the position they have in the definition of the function. (Example: `greeting("Bob", prefix="Hi")`, where you rely on `"Bob"` being in the `name` position.) These are called positional variables.

  - you can give it the explicite variable name it has in the function definition. In this case, the order of variable names does not matter. (Example: In `greeting(prefix="Welcome", name="Charlie")`, the output was correct although the order of the two variables differs from the function definition.)

- All positional arguments must come before keyword aguments.
- Both functions from above have a *docstring*, which is the multiline string after the `def`-line. They indicate what this function is doing and are displayed in various places, e.g. when you hover over a function in your jupyter notebook.
- Sometimes, it is too much to define an own function for a very simple task. In this case, Python3 comes with `lambda`-functions, which are very quick: `lambda x: x * x`. Most use cases are with container data types, and we will have one in Section 4.1.

3.5 Type hints and assertions (`assert`, type annotations)

claude generated

Python does not enforce types, but you can annotate them. This makes function signatures look like mathematical declarations (e.g. $f : \mathbb{R} \to \mathbb{R}$) and serves as documentation. The `assert` statement is a simple way to check assumptions at runtime.

- `def f(x: float) -> float:`: declare that `f` takes a float and returns a float. This is not enforced by Python, but helps readability and tools like `mypy`.
- `assert condition, "message"`: raise an `AssertionError` if `condition` is `False`.

```
def square(x: float) -> float:
    """Return x squared."""
    return x * x

print(square(3.0))
# This also works, since Python does not enforce the type hint:
print(square(5))
```

```
9.0
25
```

```
def safe_sqrt(x: float) -> float:
    """Return the square root of x, but only for non-negative x."""
    assert x >= 0, f"Expected non-negative input, got {x}"
    return x ** 0.5

print(safe_sqrt(4.0))
```

```
2.0
```

```python
# This will raise an AssertionError
print(safe_sqrt(-1.0))
```

```
AssertionError: Expected non-negative input, got -1.0
---------------------------------------------------------------------------
AssertionError                            Traceback (most recent call last)
Cell In[12], line 2
      1 # This will raise an AssertionError
----> 2 print(safe_sqrt(-1.0))
Cell In[11], line 3, in safe_sqrt(x)
      1 def safe_sqrt(x: float) -> float:
      2     """Return the square root of x, but only for non-negative x."""
----> 3     assert x >= 0, f"Expected non-negative input, got {x}"
      4     return x ** 0.5
AssertionError: Expected non-negative input, got -1.0
```

Type hints are not enforced, but they make your code much easier to read, especially when functions become more complex. The `assert` statement is useful for catching mistakes early during development. (It can be disabled in production code.)

3.6 Exception handling (`raise`, `try ... except`)

In code, various things happen, and some go wrong. Raising (throwing) errors, catching them, and dealing with them is the topic of this section. There are many error types implemented. Let us look at some of them:

Exception	When it happens	Example
`ValueError`	Right type, wrong value	`int("abc")`
`TypeError`	Wrong type used	`"2" + 3`
`IndexError`	Index out of range	`[1,2][5]`
`KeyError`	Dictionary key missing	`{"a":1}["b"]`
`ZeroDivisionError`	Division by zero	`1 / 0`
`FileNotFoundError`	File does not exist	`open("x.txt")`
`PermissionError`	No access rights	opening protected file
`AttributeError`	Object has no attribute	`"abc".foo`
`NameError`	Variable not defined	`print(x)`
`ImportError`	Import fails	`import nonexisting`
`ModuleNotFoundError`	Module not found	`import xyz`
`AssertionError`	`assert` fails	`assert False`

We obtain an error here:

```
print("This will not work")
print(f"1/0 = {1/0}")
print("If this print works, the program has continued past 1/0")
```

```
This will not work

ZeroDivisionError: division by zero
---------------------------------------------------------------------------
ZeroDivisionError                         Traceback (most recent call last)
Cell In[13], line 2
      1 print("This will not work")
----> 2 print(f"1/0 = {1/0}")
      3 print("If this print works, the program has continued past 1/0")
ZeroDivisionError: division by zero
```

However, if we *catch* the error, the code can run past it:

```
try:
    print("This will not work")
    print(f"1/0 = {1/0}")
    print("This never runs")
except ZeroDivisionError as e:
    print("You must not divide by zero!")
    print("This is the error message:", e)
print("The program continues!")
```

```
This will not work
You must not divide by zero!
This is the error message: division by zero
The program continues!
```

Now we know how to *catch* errors, but often, we need to *throw* them. This is actually simple using `raise`:

```
def compute_sth(x):
    if type(x) != int:
        raise TypeError("x must be an int!")
    y = x ** x
    return y
```

```python
try:
    print(compute_sth(2))
    print(compute_sth("abc"))
except TypeError:
    print("Calculation not possible, sorry")
```

```
4
Calculation not possible, sorry
```

3.7 Context management (`with`)

Often, our Python3 file (or notebook) must interact with other files or ressources (databases, other connections). In order to do this safely, we are using `with`, which guarantees that the ressource is entered (opened) and exited (closed) correctly. The general structure is as follows:

```
with something as name:
    do_something()
```

The simplest example is writing a file, or reading a file.

```python
with open("hello.txt", "w") as f:
    f.write("Hello")
    print("File written.")
```

```
File written.
```

```python
with open("hello.txt", "r") as f:
    text = f.read()
print("This is the content of the file:", text)
```

```
This is the content of the file: Hello
```

Here, `"hello.txt"` is closed at the end of the `with` block. The second argument of `open` comes with the following options: * `r`: read (text mode) * `w`: (over-)write (text mode) * `a`: append (text mode) * `x`: create new (i.e. throw error if already exists) * `rb`: read (binary mode) * `wb`: (over-)write (binary mode) * `ab`: append (binary mode) * `xb`: create new (i.e. throw error if already exists)

3.8 File I/O (`open`, `read`, `write`)

claude generated

Reading and writing files is one of the most common tasks in data analysis. Python's built-in `open` function, combined with the `with` statement from above, provides a clean way to handle files. Here, `f` is an open file object.

- `open(path, mode)`: open a file. The most common modes are `"r"` (read), `"w"` (write/overwrite), `"a"` (append). See the section on `with` above for the full list.
- `f.read()`: read the entire file as a single string.
- `f.readline()`: read a single line.
- `f.readlines()`: read all lines into a list of strings.
- `f.write(s)`: write string `s` to the file.
- `f.writelines(lines)`: write a list of strings to the file.

```
# writing a file
with open("example.txt", "w") as f:
    f.write("Line 1\n")
    f.write("Line 2\n")
    f.write("Line 3\n")
print("File written.")
```

```
File written.
```

```
# reading the entire file
with open("example.txt", "r") as f:
    content = f.read()
print(content)
```

```
Line 1
Line 2
Line 3
```

```
# reading line by line (useful for large files)
with open("example.txt", "r") as f:
    for line in f:
        print(line.strip())
```

```
Line 1
Line 2
Line 3
```

```python
# reading all lines into a list
with open("example.txt", "r") as f:
    lines = f.readlines()
print(lines)
```

```
['Line 1\n', 'Line 2\n', 'Line 3\n']
```

```python
# appending to a file
with open("example.txt", "a") as f:
    f.write("Line 4\n")

with open("example.txt", "r") as f:
    print(f.read())
```

```
Line 1
Line 2
Line 3
Line 4
```

When working with data, you will usually use higher-level functions like `pd.read_csv()` (see Section 7.3). But understanding basic file I/O is important for debugging and for handling non-standard file formats.

3.9 Exercises

Exercise 1 Can you code the `FizzBuzz` example in a single line of code? So, dependent on some `n`, give one line of code, which gives `Fizz` if `n` is a multiple of 3, `Buzz`, if it is a multiple of 5, and `FizzBuzz` if both conditions are satisfied.

Exercise 2 Multiple `elif`s in a row are not very elegant. Python3 (since version 3.10) comes with `match`. Find out how you can change the `FizzBuzz` example of the main text by using `match` instead of `elif`s.

Exercise 3 Write a version of the code for finding the gcd by running (not from 1 to `a`, but) from `a` to `1`, and see if the corresponding number divides both, `a` and `b`. Apparently, you can break the `for`-loop when you found the first such number. Implement that breaking of the `for`-loop.

Exercise 4 Compute $\sum_{i=1}^{100} i$ using a `for`-loop.

Exercise 5 Find the largest n such that $\sum_{i=1}^{n} i < 1000$.

Exercise 6 `for`-loops can also become nested. Compute $\sum_{i=1}^{100} \sum_{j=1}^{i} j$. (Note the double indentation for the inner `for`-loop!)

Exercise 7 Define a function computing the sum of the first `n` integers.

Exercise 8 Write a function (without using imports) which computes the sum of all digits (cross sum) of an int.

```
def check_age(age_as_string):
    try:
        age = int(age_as_string)
    except ValueError as e:
        print("Age must be an int.")
        age = -1
    return age

print(check_age(25))
print(check_age("abc"))
```

```
25
Age must be an int.
-1
```

4 Container data types

Download notebook.

When working with data, we usually not only have *one* int or *one* str, but many of them. For this situation, there are the container data types, `list`, `tuple`, `set`, and `dict`.

4.1 Lists (`list`)

The notation for a list comes with square brackets, where the entries are separated by a comma. The list of numbers 1, 2, 3 is `[1,2,3]`. A list of names is e.g. `["Alice", "Bob", "Charlie", "Donna"]`. So, for a `list`, the entries have simple (or again container) data types.

- We can access the entries of a list using square brackets. However, we have to be careful if the length of the list is too short.
- We can access sublists by using `i:j` for the entries `i,...,j-1`. If we want to have the beginning until `j-1`, we use `:j`. If we want everything after `i`, we use `i:`.
- Elements of a list can be lists (or other container data types) themselves.
- The types of elements of a list can be mixed.
- When talking about `for`-loops, we already had `range(n)`, which stands for `0,...,n-1`. This is not exactly a list, but can be turned into one by `list(range(n))`. (More precisely, there is also `range(start, stop, step)` for `start, start + step, start + 2*step,...`, which stops at `stop-1` at the latest.)

```
# a list of ints
x = [1,2,3]
print(x[0])
print(x[3] if len(x) > 3 else None)
```

```
1
None
```

Slicing refers to sublists.

```
x = [1,2,3]
print(f"The first and second element: {x[0:2]}")
print(f"The first two elements: {x[:2]}")
print(f"All elements after the first element: {x[1:]}")
```

```
The first and second element: [1, 2]
The first two elements: [1, 2]
All elements after the first element: [2, 3]
```

Playing around with `range`.

```
print(f"The first and second odd number: {list(range(1, 5, 2))}")
print(f"The first and second odd number: {list(range(1, 100, 2))[:2]}")
```

```
The first and second odd number: [1, 3]
The first and second odd number: [1, 3]
```

The elements of a list can be lists themselves.

```
y = [[1,2], [3,4,5], 6, 7, 8]
print(y[1])
print(y[1][0])
```

```
[3, 4, 5]
3
```

A list can have mixed types.

```
z = [1, "two", [3,4], 5.0]
print(z[1])
```

```
two
```

Here are some functions on lists.

- `len(l)`: length of `l`.
- `sum(l)`: sums all elements of `l`, if possible.
- `min(l)`, `max(l)`: computes the minimum/maximum of `l`.
- `sorted(l)`: returns a sorted version of `l`.

- `filter(fun, l)`: returns a filtered list, which only contains elements `x` of `l` with `fun x == True`.
- `list.append(l, x)` (or `l.append(x)`) (where `x` is any object) appends the element `x` to the list `l`.
- `list.extend(l, x)` (or `l.extend(x)`) (where `x` is a list) extends `l` by all elements of `x`. (You can also write `l + x` for this.)
- `list.insert(l, i, x)` (or `l.insert(i,x)`) inserts `x` at position `min(i, len(l))` in `l`.
- `list.remove(l, x)` (or `l.remove(x)`) removes the first occurrence of `x` in `l`.
- `list.pop(l)` (or `l.pop()`) removes the last element from `l`.
- `list.index(l, x)` (or `l.index(x)`) gives the index of the first occurrence of `x` in `l`.
- `list.count(x)` (or `l.count(x)`) counts the number of occurrences of `x` in `l`.
- `list.sort(l)` (or `l.sort()`) sorts the list.
- `list.reverse(l)` (or `l.reverse()`) reverses the list.

For the functions starting with `list`, there is a catch: While we mentioned that a function takes its input (here the list), and does not alter it, `list`s are mutable objects, which means that they are changed when calling such a function on them. So, e.g. after `l.sort()`, the list `l` itself is sorted, and there is no return value of that function (more precisely, the return value is `None`.) This can be counterintuitive:

```
l = [3,2,1]
# This gives none since the return value of l.sort() is None
print(l.sort())

l = [3,2,1]
# This changes l and makes it sorted.
l.sort()
# Now we can print it
print(f"l = {l}")

# In contrast, sorted() does not change the list but returns a new sorted list.
l = [3,2,1]
print(sorted(l))
```

```
None
l = [1, 2, 3]
[1, 2, 3]
```

Assume you want to sort a `list` not as usual. In this situation, you will find out about the `key`-parameter in `sorted`. It takes a function, we sorts the list according to the values of this function. As an example, assume we want to sort a list of `int`s according to their abolute value:

```
l = [5, -4, 0, 7, -3, 2]
print(sorted(l, key = lambda x: abs(x)))
```

```
[0, 2, -3, -4, 5, 7]
```

Similarly, you can e.g. filter the list by only taking numbers which are at least one:

```
l = [5, -4, 0, 7, -3, 2]
print(list(filter(lambda x: x >= 1, l)))
```

```
[5, 7, 2]
```

Let us shortly describe two functions on a list of `bools` `l`.

- `all(l)` gives `True` if all entries of `l` are 'True.
- `any(l)` gives `True` if at least one entry of `l` is 'True.

4.2 Tuples (`tuple`)

At first glance, a `tuple` is like a `list`, but comes with normal brackets `()` instead of square brackets `[]`. So, `(1,2,3)` is the tuple of these three numbers, and `("Alice", "Bob", "Charlie") is a tuple of three names. For the tuple` l, `still,`l[0]`gives the first entry etc. The main difference between`tuple`and`list`is that`lists `are more flexible when adding or changing entries, and`tuple's are fixed (immutable) once they have been created.

```
l = ("Alice", "Bob", "Charlie")
# The next line does not work, since l is immutable
# l[0] = "Anna"
l = ["Alice", "Bob", "Charlie"]
# The next line works, since l is mutable.
l[0] = "Anna"
print(f"l = {l}")
```

```
l = ['Anna', 'Bob', 'Charlie']
```

An important aspect of tuples is that single values can easily be packed into a tuple. An important example is to use a tuple in order to assign two or more values at the same time:

```python
def min_max(a,b):
    (mi, ma) = (a if a < b else b, b if a < b else a)
    return mi, ma

a = 20
b = 10
res = min_max(a,b)
print(f"Minimum and maximum are {res[0]} and {res[1]}.")
```

```
Minimum and maximum are 10 and 20.
```

A special note applies to tuples of length 1. Here, note that e.g. `(1)` does not create a `tuple` of length 1, but an `int`. Instead, use `(1,)` for the `tuple` of length one.

```python
thisisnotuple = (1)
print(type(thisisnotuple))
print(thisisnotuple[0])
thisisatuple = (1,)
print(type(thisisatuple))
print(f"Its first and only element is {thisisatuple[0]}.")
```

```
<class 'int'>

TypeError: 'int' object is not subscriptable
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[11], line 3
      1 thisisnotuple = (1)
      2 print(type(thisisnotuple))
----> 3 print(thisisnotuple[0])
      4 thisisatuple = (1,)
      5 print(type(thisisatuple))
TypeError: 'int' object is not subscriptable
```

4.3 Sets (`set`)

As in mathematics, sets have no special order, and there are noduplicates within a set. In Python3, they can be created using curly brackets, so e.g. `{1,2,3}` as well as `{1,3,2,3,1}` is the set of the numbers `1,2,3`. Although there are functions for updating sets (as for lists), the

most frequent application is for removing duplicates. For example, if you have a `list`, which can contain duplicates, `list(set(l))` is the same list (maybe the order changed) but with duplicates removed.

Here are some functions on some `s` and `t`: * `len(s)`: returns the number of unique elements in `s`. * `s | t`: The union of `s` and `t`. * `s & t`: The intersection of `s` and `t`. * `s - t`: The difference of `s` and `t`.

```
l = [1,3,2,4,3,2]
print(f"There are {len(set(l))} unique elements in {l}.")
```

Unions, intersection, and set differences work as expected:

```
s = {1,2,3}
t = {5,4,3}
print(f"The union of {s} and {t} is {s | t}.")
print(f"The intersection of {s} and {t} is {s & t}.")
print(f"The difference of {s} and {t} is {s - t}.")
```

4.4 Dictioanries (`dict`)

Dictionaries (`dict`) are nothing but a collection of key-value pairs. Here is an example:

```
d = {
        "first_name": "Alice",
        "last_name": "Müller",
        "age": 29,
        "hobbies": ["reading", "cycling", "chess"]
    }
print(f"""{d['first_name']} {d['last_name']} is {d['age']} and likes
    {', '.join(d['hobbies'])}.""")
```

```
Alice Müller is 29 and likes
    reading, cycling, chess.
```

- As you see in the example, the f-string uses nested quotation marks, `"...'...'..."`. For this reason, we use single quotation marks `''` as inner separators and double quotation marks `""` for the outer quotation marks. If there is no nesting, there is no difference in using single/double quotation marks.
- keys in a dict can be any immutable data type (like `int`, `str`, `float`, `tuple`).

As you see, accessing field `'age'` in the `dict p` is done using `p['age']`. However, assume `'age'` is not a field in `p`. Then, `p['age']` gives an error. For this reason, it is often better to use the `get`-function:

```
print(f"{d['first_name']} {d['last_name']} lives in {d.get('address')}.")
# This version uses a default field in get
print(f"""{d['first_name']} {d['last_name']} lives
    in {d.get('address', 'unknown')}.""")
# We can also add "address" to the dict
d["address"] = "Freiburg, Germany"
print(f"""{d['first_name']} {d['last_name']} lives
    in {d.get('address', 'unknown')}.""")
```

```
Alice Müller lives in None.
Alice Müller lives
    in unknown.
Alice Müller lives
    in Freiburg, Germany.
```

Some of the most important functions on `d` as a `dicts` are: * `dict.keys(d)` (or `d.keys()`): Returns the list of all keys. * `dict.values(d)` (or `d.values()`): Returns the list of all values. * `dict.items(d)` (or `d.items()`): Returns the list of tuples (key, value).

```
print(f"The keys are {', '.join(d.keys())}.")
print(f"The values are:")
for v in d.values():
    print(str(v))
print(f"The key-value-pairs are:")
for key, value in d.items():
    print((key, str(value)))
```

```
The keys are first_name, last_name, age, hobbies, address.
The values are:
Alice
Müller
29
['reading', 'cycling', 'chess']
Freiburg, Germany
The key-value-pairs are:
('first_name', 'Alice')
('last_name', 'Müller')
```

```
('age', '29')
('hobbies', "['reading', 'cycling', 'chess']")
('address', 'Freiburg, Germany')
```

Let us mention some more functions on `dicts`:

- `dict.update(d, d')` (or `d.update(d')`): extends `d` by the keys in `d'`, and overwrites entries in `d` by the values of `d'`. Another possibility is `d |= d'` (similar to a set-union).
- `dict.pop(d, key, default)` (or `d.pop(key, default)`): removes the key `key` from `d` and returs its value (or `default`, if the key isn't found).
- `del d[key]`: Deletes the entry with key `key` from `d`.

```python
d |= {
    "profession" : "student",
    "subject" : "Mathematics"
}
del d["age"]
for key, value in d.items():
    print((key, str(value)))
```

```
('first_name', 'Alice')
('last_name', 'Müller')
('hobbies', "['reading', 'cycling', 'chess']")
('address', 'Freiburg, Germany')
('profession', 'student')
('subject', 'Mathematics')
```

Here is a last example for a list of dicts.

```python
people = [
    {
        "first_name": "Alice",
        "last_name": "Müller",
        "age": 29,
        "hobbies": ["reading", "cycling", "chess"]
    },
    {
        "first_name": "Bob",
        "last_name": "Schmidt",
        "age": 35,
        "hobbies": ["hiking", "photography", "cooking"]
```

```
    },
    {
        "first_name": "Clara",
        "last_name": "Weber",
        "age": 22,
        "hobbies": ["painting", "piano", "running"]
    },
    {
        "first_name": "David",
        "last_name": "Fischer",
        "age": 41,
        "hobbies": ["gardening", "woodworking", "cycling"]
    }
]
for p in people:
    print(f"{p["first_name"]} {p["last_name"]} is {p["age"]} and likes "
        + ", ".join(p["hobbies"]) + ".")
```

```
Alice Müller is 29 and likes reading, cycling, chess.
Bob Schmidt is 35 and likes hiking, photography, cooking.
Clara Weber is 22 and likes painting, piano, running.
David Fischer is 41 and likes gardening, woodworking, cycling.
```

4.5 Comprehensions

Python3 has very expressive (i.e. short) ways to alter container data types. The general form is (where "iterable" is a container data type):

```
[expression for variable in list if condition]
[expression_if_true if condition else expression_if_false for variable in list]
{key: value for item in dict if condition}
```

Let us make some examples, and you will see how powerful these complehensions are!

```
# The squares of all even numbers below 10
print([x*x for x in range(10) if x % 2 == 0])
# The names of people who like cycling
print("People who like cycling are " +
    ', '.join([f'{p["first_name"]} {p["last_name"]}' for p in people
        if "cycling" in p['hobbies']]) + ".")
```

```
[0, 4, 16, 36, 64]
People who like cycling are Alice Müller, David Fischer.
```

4.6 Exercises

Exercise 1 For two strings `s` and `t`, write a function `myCount(s, t)`, which counts the numbers of occurrences of `t` in `s`. Unlike `count`, it should count the number of positions in `s` starting with `t`. For example, `"bababab".count("bab")` gives 2, but `"bababab".myCount("bab")` should give 3.

Exercise 2 Find out how you can sort a list descendingly.

Exercise 3 Write a function which removes duplicates from a list (you might want to use `set` here), but keeps the order of the list.

Exercise 4 Given a list of names [“ALICE”, “BOB”, “CHARLIE”], convert them all to lowercase.

Exercise 5 From a list of fruits [“apple”, “banana”, “kiwi”, “mango”, “pear”], create a list of fruits that have exactly 5 letters.

Exercise 6 Create a list of tuples representing all possible `(x, y)` coordinates where `x` is from `[1, 2, 3]` and `y` is from `[4, 5]`.

Exercise 7 Given the dict `{"a": 1, "b": 2, "c": 3}`, create a new dictionary that swaps keys and values `{1: "a", ...}`. This is best done using a comprehension.

Exercise 8 Assume you have `a = [10, 20, 30, 40]` and `mask = [True, False, True, False]`, you might want obtain a sublist of `a` which only contains the positions where `mask` has `True. Find out (using AI) how this works (without using other libraries), and describe what your code does. (You will probably find out about the`zip‘-function, which you should look at more closely.)

5 Some useful libraries/packages/modules

Download notebook.

In this chapter, we take a little break. Although there are lots of things to learn, the material covered here has the role of helping to learn about the Python3 ecosystem and what is possible, rather than learn the details of various labraries. The first section, however, is central and will be needed in any Python3 project you will start.

5.1 Loading modules (`import`)

As in all programming languages, not all python programs come in a single file. If a Python3 project spans multiple files, an `import` becomes necessary. In Python3, there is syntax for `import`s as follows:

Syntax	Meaning
`import module`	full module
`import module as name`	alias
`from module import x`	specific objects
`from module import *`	everything (avoid)
`import module.submodule`	submodule

Generally, there are packages (folders), modules (a `.py`-file within a package, or a single `py`-file), and submodules. Every submodule is a module. In any case, the import loads the (sub-)module, and executes all top-level code inside it (i.e. everything which is not inside a function).

There are three cases:

1. You wrote some code in file `one.py` (e.g. some functions) which you want to use in file `two.py`. (Note, btw, that the first digit in `one.py` must be a letter.) Assume `one.py` is as follows:

```
# File: one.py

def myConst():
    return 42
```

Then, the import in `two.py` is either `import one` (both files must be in the same folder), or `from one import myConst` or (not recommended) `from one import *`. In the first case, you must prefix `myconst()` by `one`:

```
# File: two.py

import one
print(f"My constant is {one.myConst()}.")
```

In the second case, there is no prefix:

```
from one import myConst
# or
from one import *
print(f"My constant is {myConst()}.")
```

Observe that the latter option comes with the disadvantage that you cannot see from the call of `myconst()`, if it is defined in `two.py` or in`one.py`, so it becomes harder to keep track of your code if things become more complicated.

If you have a folder `myPackage`, which holds `one.py`, you can must write `import myPackage`, `import myPackage.one` or `import myPackage.one as myNameOne` with differing ways to use `myConst()`:

```
import myPackage
print(f"My constant is {myPackage.one.myConst()}.")

import myPackage.one
print(f"My constant is {one.myConst()}.")

import myPackage.one as myName
print(f"My constant is {myName.myConst()}.")
```

2. You want to use functions from a built-in library. Then, a simple `import module` is often the best choice. We will see examples in Section 5.2.
3. You want to use functions from a library, which needs to be installed. Here, before the `import module` even works, you need to install the module in your python environment. In a terminal, within your Python3 environment, use

```
# Uncomment the next line and run the cell for installation.
# This only needs to be done once for every package.
# If pip3 does not work, use pip instead.
# !pip3 install numpy
```

and then, `import numpy` will work. We will see examples for this e.g. in **?@sec-loading**.

5.2 Mathematics (`math`)

The mathematical library in Python is used by `import math`. Then, you have e.g. the following functions, which are self-explanatory: `math.sqrt()`, `math.exp()`, `math.log()`, `math.sin()`, `math.cos()`, and constants like `math.pi` and `math.e`.

5.3 Dates and times of day (`datetime`)

In data science, we often have to handle dates. Their disadvantage is that there is no standard way to write these as strings. In Python3, this and various other problems around dates and times is solved by the `datetime`-library.

Submodules (in fact, these are `class`es in this case) of this package are

- `date`: handling dates
- `time`: handling times
- `datetime.datetime`: handling `datetimes`, i.e. dates and times together
- `timedelta`: time differences
- `tzinfo` and `timezone`: handling timezones

We will here use

```
import datetime
```

at the cost that `datetime`-objects are only accessible by saying `datetime.datetime` (the first `datetime` being the package, the second `datetime` being the class/submodule).

Let us create a `datetime`-object‘:

```
dt = datetime.datetime(2026, 4, 21, 12, 15)
print(dt)
print(f"year: {dt.year}")
print(f"month: {dt.month}")
print(f"day: {dt.day}")
print(f"hour: {dt.hour}")
print(f"minutes: {dt.minute}")
print(f"second: {dt.second}")
print(f"microsecond: {dt.microsecond}")
print(f"date: {dt.date()}")
print(f"time: {dt.time()}")
```

```
2026-04-21 12:15:00
year: 2026
month: 4
day: 21
hour: 12
minutes: 15
second: 0
microsecond: 0
date: 2026-04-21
time: 12:15:00
```

which is the starting time of the first session in this course. As we see here, we can extract year,..., microseconds from the `datetime`-object.

Often, we are given strings such as `"21.4.26, 12:15"`, and want to convert this into a `datetime` object. For this, we have `datetime.datetime.strptime`:

```
s = "21.4.26, 12:15"
dt = datetime.datetime.strptime(s, "%y.%m.%d, %H:%M")
print(s + f" is converted to {dt}.")
```

```
21.4.26, 12:15 is converted to 2021-04-26 12:15:00.
```

(For the correct way to place %, y, m, d etc., it is in fact best to consult your favourite AI.)

Reversely, we want to print a `datetime`-object in a certain way. For this, we have `datetime.datetime.strftime`:

```
print(dt.strftime("%-m/%-d/%y, %H:%M"))
```

```
4/26/21, 12:15
```

which is the US-style output.

Next, assume we want to compute the same time 12 weeks from now. For this, we use `datetime.timedelta`.

```
print(dt + datetime.timedelta(days=84))
```

```
2021-07-19 12:15:00
```

Here are some more remarks:

- `datetime.date` is not used very often. The reason is that you can recover a `date` from a `datetime`, but not the other way round.
- `datetime.time` has the same limitations.
- We will not cover time-zones here, but if you have global data, they will become important!

5.4 Decimal numbers (`decimal`)

We already saw that:

```
print(f"Rounding error: 0.2 + 0.1 = {0.2 + 0.1}.")
```

```
Rounding error: 0.2 + 0.1 = 0.30000000000000004.
```

The reason is that computers in general use a base of 2 for all computations, and 0.3 has a bad representation in this system. While this small error is acceptable often, it is not in some contexts (above all, finance). In this case, we can use the `decimal` library. Within `decimal`, a base of 10 is used, but calculations are slower than for `floats`.

```
from decimal import Decimal
print(f"Exact calculation: 0.2 + 0.1 = {Decimal('0.2') + Decimal('0.1')}.")
```

```
Exact calculation: 0.2 + 0.1 = 0.3.
```

5.5 Regular expressions (`re`)

Sometimes, you need to find patterns in a string. In data science, this happens if some field contains unstructured text, and you want to extract information. As an example, assume you load data from a csv-file, and one columns contains all email-adresses of the corresponding item. If you read this, you have something like `"muller@gmail.com; muller@yahoo.com"`. However, you want to separate the two addresses, and store them in a `list`. One way to approach this is using a regular expression (which in this case reads `"\S+@\S+\.\S+"`):

```
import re

text = "muller@gmail.com; muller@yahoo.com"

emails = re.findall(r"\S+@\S+\.\S+", text)
print(emails)
```

```
['muller@gmail.com;', 'muller@yahoo.com']
```

Let us quickly break this down to some extent, looking at `"\S+@\S+\.\S+"`: * `\S`: any non-whitespace character * `+`: one or more of the previous thing * `@`: exactly the `@` symbol * `\.`: exactly the dot `.` character (separating e.g. gmail from com)

We can agree that this exactly what an email address looks like. The `re.finall()` finds all places in `text` which match this pattern and extracts them, putting them into a list. Note that there are more patterns to find than `\S`, `+` and exact matches. For this reason, it is advised to use AI when it comes to regular expressions.

5.6 Storing and restoring objects (`pickle`)

Sometimes, you want to store intermediate objects. One way is to write them out in files, but you can also use the `pickle`-library in order to write Python3 objects to disc. Here is an example:

```
import pickle

data = {"name": "Alice", "age": 30}

# Save to file (pickle)
with open("data.pkl", "wb") as f:
    pickle.dump(data, f)
```

```
# Load from file (unpickle)
with open("data.pkl", "rb") as f:
    loaded_data = pickle.load(f)

print(loaded_data)
```

```
{'name': 'Alice', 'age': 30}
```

5.7 Testing code (pytest)

```
# Uncomment the next line and run the cell for installation.
# If pip3 does not work, use pip instead.
# !pip3 install pytest
```

When starting to code, producing reliable programs is both important and difficult. One – not the best – way to approach this is to rely on the program until something breaks. It is actually better to *test* the code, i.e. to write tests. This means that you tell your program that you expect a certain behavior.

A simple and powerful library for this is `pytest`. When working with `.py`-files, you usually separate such tests in a separate file. Every test-case is within a function with name starting with `test_`. In a terminal, you can run `pytest` on that file, which performs all tests, and reports any error.

However, we are working with Jupyter notebooks here, so we cannot run `pytest` on separate files. Although there are workarounds (the package `ipytest`), we will only go through some basics here, without importing `pytest`. This means we are introducing `assert`.

```
x = 5
y = -1
assert 2 + 2 == 4
assert 2 + 2 == 5
```

```
AssertionError:
---------------------------------------------------------------------------
AssertionError                            Traceback (most recent call last)
Cell In[12], line 4
      2 y = -1
      3 assert 2 + 2 == 4
----> 4 assert 2 + 2 == 5
AssertionError:
```

5.8 Using the operating system (os)

For the kind of projects we envision, we often must access files, which happens via the operating system. We might also want to see all files in some directory, access them one by one, etc. Here, `folder` is the string of the name of a folder, and `file` is the string for the name of a file. * `os.getcwd()`: returns the current working directory. * `os.chdir(folder)`: this is a Python3 command for `cd`. * `os.path.join(folder", file)`: gives a complete path from the folder and file name. * `os.listdir(folder)`: returns a list of files within the folder. * `os.path.exists(file)`: returns true or false if the file exists or not. * `os.walk(folder)`: gives the triple `(root, dirs, files)`, where `root` is the absolute path of the folder, `dirs` are all direct subfolders in that folder, and `files` are the files in that folder. If folder has subfolders, it returns a list of triples for all subfolders.

```
import os
# Print all ipynb-files in the current folder
for file in os.listdir("."):
    if file.endswith(".ipynb"):
        print(file)
```

```
# Print folders and files (only first 5 entries of os.walk)
for i, (root, dirs, files) in enumerate(os.walk(".")):
    if i >= 5:
        print("... (truncated)")
        break
    print("ROOT:", root)
    print("DIRS:", dirs[:5], "..." if len(dirs) > 5 else "")
    print("FILES:", files[:5], "..." if len(files) > 5 else "")
    print()
```

```
ROOT: .
DIRS: ['myFiles', '07_handling_data_files', 'archive', '08_analyzing_data_files', '.quarto']
FILES: ['04_container_data_types.quarto_ipynb', 'hello.txt', '07_handling_data.quarto_ipynb'

ROOT: ./myFiles
DIRS: ['myPackage']
FILES: ['01.ipynb', 'two.py', 'one.py']

ROOT: ./myFiles/myPackage
DIRS: ['__pycache__']
FILES: ['one.py']
```

```
ROOT: ./myFiles/myPackage/__pycache__
DIRS: []
FILES: ['one.cpython-313.pyc']

ROOT: ./07_handling_data_files
DIRS: ['figure-pdf']
FILES: []

... (truncated)
```

5.9 Starting other programs (subprocess)

Sometimes, you want to start some piece of software (a command line tool) from within your Python3 program, and collect the result. You could use a script for this, which starts the command line tool, and rus your Python3 script afterwards, but you can as well use Python3 for the whole setting. We only give a basic example:

```
import subprocess
filename = "file.tex"
try:
    result = subprocess.run(
        ["pdflatex", "-interaction=nonstopmode", filename],
        capture_output=True, # capture stdout and stderr output instead of printing
        text=True, # returns output as string, not binary
        check=True # raises an error if the command fails
    )
    print(result.stdout)
except FileNotFoundError:
    print(f"File {filename} not found.")
```

5.10 Exercises

Exercise 1 Apparently, `"muller@gmail.com; wrong-email; muller@yahoo.com"` only contains two valid email addresses. Use AI and `re` in order to produce and understand code, which separates the correct from the incorrect email addresses.

Exercise 2 A bank manager has an idea: Interests should be computed daily (as usual), but rounded to cents (round up if last digit is 5 or above) at the end of the day. (So, sub-cents are not kept for computing the account balance.) Assume someone has invested 1 Euro on January 1st, 0 (birth of christ) with an interest rate of 5% per year. * Compute the account

balance *as usual*, i.e. keeping sub-cents at all days. * Compute the account balance using the above strategy, i.e. ignoring sub-cents at the end of each day.

Exercise 3 Instead of `os.walk()`, we can also use the `pathlib` library. Find out how the equivalent of `os.walk()` works when using `pathlib`.

Exercise 4 Create some `txt`-files in the current working directory. Then, write a Pathon3 program that iterates through all such files, creates `all.txt`, where the content of all you `txt`files is appended. If `all.txt` already exists, the program should give a warning that this file is overridden.

Exercise 5 What is your age in seconds/minutes/hours/days/months?

Exercise 6 Below are some sales dates and quantities.

- Write a function, with `start` and `end` as `datetime` as input, which gives you the total amount of sales during this time. (You will need to convert `date` as given in the data to `datetime`. Note that `<`, `>`, `<=`, `>=`, `==` work for `datetime` as expected.)
- Write a function, with a date `date` and a number days `days` as input, which returns the average amount of sales per day during the last `days` since `date`.

```
data = [
        ("2026-01-01", 120),
        ("2026-01-02", 135),
        ("2026-01-03", 128),
        ("2026-01-08", 150),
        ("2026-02-01", 210),
        ("2026-02-03", 215)]
```

6 Numerical mathematics

Download notebook.

We will be using `numpy` {#sec-numpy} and some `scipy`. These packages provide powerful and fast computations for numerical mathematics. (Most of it is implemented in `C`.) We will only learn about some parts of it, some linear algebra (using `numpy`), optimization (using `scipy`) and statistics (using `scipy`). Together with `pandas` (see Section 7.3) for data handling, this is the basic framework for data analysis in Python3.

6.1 Arrays, vectors, matrices (`np.array`)

Let us start by importing `numpy`.

```
# uncomment the following line for installing numpy
# !pip3 install numpy
import numpy as np
```

(Then, the functions from `numpy` are called `np.something`.)

A vector or matrix is an `np.array`. `numpy` interprets a vector as a column vector. Here, `x` is an `np.array` of type `float64`:

- `x.ndim`: number of dimensions of `x`.
- `x.shape`: tuple of length `x.ndim` which gives the length of dimension `i` as entry `i`. (Recall that e.g. `(3,)` is a tuple of length `1`.)
- `x.dtype`: type of entries in `x`.
- `x.reshape(m, n)`: returns a view of `x` with shape `(m, n)`. The total number of entries must match. Using `-1` for one dimension lets NumPy infer it.
- `x.T`: the transpose of `x`. For a 1d array, this has no effect.
- `x.copy()`: returns an independent copy of `x`.
- `x.flatten()`: returns a 1d copy of `x`.

```python
x = np.array([1.0, 2.0, 3.0])
A = np.array([
    [1.0, 0.0, 2.0],
    [0.0, 1.0, 1.0]
])
print("x =", x)
print("A =")
print(A)
```

```
x = [1. 2. 3.]
A =
[[1. 0. 2.]
 [0. 1. 1.]]
```

```python
print("x.shape =", x.shape)
print("A.shape =", A.shape)
print("x.ndim =", x.ndim)
print("A.ndim =", A.ndim)
print("x.dtype =", x.dtype)
```

```
x.shape = (3,)
A.shape = (2, 3)
x.ndim = 1
A.ndim = 2
x.dtype = float64
```

```python
# reshape and transpose return views, not copies
print("A.reshape(3,2) =")
print(A.reshape(3,2))
print("A.reshape(-1) =", A.reshape(-1))
print("A.T =")
print(A.T)
```

```
A.reshape(3,2) =
[[1. 0.]
 [2. 0.]
 [1. 1.]]
A.reshape(-1) = [1. 0. 2. 0. 1. 1.]
A.T =
[[1. 0.]
```

```
 [0. 1.]
 [2. 1.]]
```

```
# column vector vs row vector vs 1d array
x_col = x.reshape(-1, 1)
x_row = x.reshape(1, -1)
print("x.shape =", x.shape)
print("x_col.shape =", x_col.shape)
print("x_row.shape =", x_row.shape)
```

```
x.shape = (3,)
x_col.shape = (3, 1)
x_row.shape = (1, 3)
```

An important observation is that the shape of a vector in NumPy is `(3,)`, not `(3, 1)`. This is mathematically convenient in some places and dangerous in others. Much of good NumPy programming consists in keeping track of such distinctions.

6.2 Creating arrays (`np.zeros`, `np.ones`, `np.arange`, `np.linspace`, `np.eye`)

There are several convenient ways to create arrays:

- `np.zeros(n)`: array of `n` zeros.
- `np.ones(n)`: array of `n` ones.
- `np.arange(n)`: array `[0, 1, ..., n-1]`. Also `np.arange(start, stop, step)`.
- `np.linspace(a, b, n)`: `n` equally spaced points from `a` to `b`.
- `np.eye(n)`: the $n \times n$ identity matrix.
- `np.empty(n)`: uninitialized array of length `n` (fast, but entries are garbage).
- `np.column_stack([a, b, ...])`: stack 1d arrays as columns of a 2d array.
- `np.vstack([A, B])` / `np.hstack([A, B])`: stack arrays vertically / horizontally.

```
print("np.zeros(4) =", np.zeros(4))
print("np.ones(3) =", np.ones(3))
print("np.arange(10) =", np.arange(10))
print("np.linspace(0, 1, 5) =", np.linspace(0, 1, 5))
```

```
np.zeros(4) = [0. 0. 0. 0.]
np.ones(3) = [1. 1. 1.]
np.arange(10) = [0 1 2 3 4 5 6 7 8 9]
np.linspace(0, 1, 5) = [0.   0.25 0.5  0.75 1.  ]
```

```
print("np.eye(3) =")
print(np.eye(3))
```

```
np.eye(3) =
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

```
# stacking arrays
a = np.array([1.0, 2.0, 3.0])
b = np.array([10.0, 20.0, 30.0])
print("np.column_stack([a, b]) =")
print(np.column_stack([a, b]))
```

```
np.column_stack([a, b]) =
[[ 1. 10.]
 [ 2. 20.]
 [ 3. 30.]]
```

6.3 Basic arithmetic

Arithmetic operations on arrays are **elementwise**. This is different from matrix multiplication.

- `x + y`, `x - y`: elementwise addition / subtraction.
- `x * y`: elementwise multiplication (not matrix multiplication!).
- `x / y`: elementwise division.
- `x ** n`: elementwise power.
- `x @ y`: inner product of vectors, or matrix multiplication for 2d arrays.

```
x = np.array([1.0, -2.0, 4.0])
y = np.array([3.0, 5.0, -1.0])
print("x + y =", x + y)
print("x * y =", x * y)
print("2 * x =", 2 * x)
print("x ** 2 =", x ** 2)
```

```
x + y = [4. 3. 3.]
x * y = [  3. -10.  -4.]
2 * x = [ 2. -4.  8.]
x ** 2 = [ 1.  4. 16.]
```

6.4 Indexing, slicing, views, copies

Indexing and slicing work similarly to lists, but NumPy arrays support multi-dimensional indexing. An important difference: slicing an array creates a **view**, not a copy. Modifying the slice modifies the original.

- `a[i]`: element at index `i`.
- `a[i:j]`: slice from `i` to `j-1`.
- `a[::k]`: every `k`-th element.
- `A[i, j]`: element at row `i`, column `j`.
- `A[:, j]`: column `j`.
- `A[i, :]`: row `i`.
- `a[mask]`: elements where boolean array `mask` is `True`.

```
a = np.arange(10)
print("a =", a)
print("a[2:7] =", a[2:7])
print("a[::2] =", a[::2])
```

```
a = [0 1 2 3 4 5 6 7 8 9]
a[2:7] = [2 3 4 5 6]
a[::2] = [0 2 4 6 8]
```

```
A = np.arange(1, 13).reshape(3, 4)
print("A =")
print(A)
print("A[:, 1] =", A[:, 1])
print("A[1, :] =", A[1, :])
print("A[:2, 1:3] =")
print(A[:2, 1:3])
```

```
A =
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
A[:, 1] = [ 2  6 10]
A[1, :] = [5 6 7 8]
A[:2, 1:3] =
[[2 3]
 [6 7]]
```

```
# views vs copies: modifying a slice modifies the original!
a = np.arange(10)
b = a[2:6]
b[0] = 99
print("a =", a)
```

```
a = [ 0  1 99  3  4  5  6  7  8  9]
```

```
# to avoid this, use .copy()
a = np.arange(10)
c = a[2:6].copy()
c[0] = -7
print("a =", a)
print("c =", c)
```

```
a = [0 1 2 3 4 5 6 7 8 9]
c = [-7  3  4  5]
```

```
# boolean masks
x = np.array([3.0, -1.0, 2.5, -4.0, 0.0, 7.0])
mask = x > 0
print("x[mask] =", x[mask])
```

```
x[mask] = [3.  2.5 7. ]
```

6.5 Vectorization and broadcasting

In numerical computing, we want to avoid Python loops whenever possible and instead work with full arrays at once. This is called **vectorization**. It is both faster and more readable.

```
x = np.linspace(0.0, 1.0, 6)
# vectorized: looks like the mathematical formula
y = 3 * x + 2
print("y =", y)
```

```
y = [2.  2.6 3.2 3.8 4.4 5. ]
```

Broadcasting allows NumPy to combine arrays of compatible but different shapes. The smaller array is "broadcast" across the larger one.

```
X = np.array([
    [1.0, 2.0],
    [3.0, 4.0],
    [5.0, 6.0]
])
b = np.array([10.0, 20.0])
# b is broadcast across all rows of X
print("X + b =")
print(X + b)
```

```
X + b =
[[11. 22.]
 [13. 24.]
 [15. 26.]]
```

```
# pairwise operations via broadcasting
x = np.array([0.0, 1.5, 3.0, 5.0])
D = np.abs(x[:, None] - x[None, :])
print("distance matrix =")
print(D)
```

```
distance matrix =
[[0.  1.5 3.  5. ]
 [1.5 0.  1.5 3.5]
 [3.  1.5 0.  2. ]
 [5.  3.5 2.  0. ]]
```

The expression `x[:, None]` turns `x` into a column vector (shape `(4,1)`), and `x[None, :]` into a row vector (shape `(1,4)`). Broadcasting then produces the full 4×4 distance matrix without any loop.

6.6 Linear algebra (`np.linalg`)

NumPy provides a comprehensive linear algebra module. Here, `x` and `y` are vectors, `A` and `B` are matrices.

- `x @ y` or `np.dot(x, y)`: inner product $\langle x, y \rangle = \sum_j x_j y_j$.

- `A @ x`: matrix-vector product.
- `A @ B`: matrix-matrix product.
- `np.linalg.norm(x)`: Euclidean norm $\|x\|_2$.
- `np.linalg.solve(A, b)`: solve $Ax = b$. Preferred over computing $A^{-1}b$.
- `np.linalg.inv(A)`: matrix inverse. Usually `solve` is preferred.
- `np.linalg.det(A)`: determinant.
- `np.linalg.cond(A)`: condition number. Large values indicate numerical instability.
- `np.linalg.eig(A)`: eigenvalues and eigenvectors.
- `np.linalg.svd(A)`: singular value decomposition.

```
x = np.array([1.0, -2.0, 2.0])
y = np.array([3.0, 1.0, -1.0])
print("<x,y> =", x @ y)
print("||x||_2 =", np.linalg.norm(x))
print("||x||_1 =", np.sum(np.abs(x)))
print("||x||_inf =", np.max(np.abs(x)))
```

```
<x,y> = -1.0
||x||_2 = 3.0
||x||_1 = 5.0
||x||_inf = 2.0
```

```
A = np.array([[1.0, 2.0], [3.0, 4.0]])
x = np.array([1.0, -1.0])
print("A @ x =", A @ x)
```

```
A @ x = [-1. -1.]
```

```
# solving Ax = b
A = np.array([[3.0, 1.0], [1.0, 2.0]])
b = np.array([9.0, 8.0])
x = np.linalg.solve(A, b)
print("solution x =", x)
print("A @ x =", A @ x)
```

```
solution x = [2. 3.]
A @ x = [9. 8.]
```

```
# eigenvalues
A = np.array([[2.0, 1.0], [1.0, 2.0]])
eigenvalues, eigenvectors = np.linalg.eig(A)
print("eigenvalues =", eigenvalues)
```

```
eigenvalues = [3. 1.]
```

One often sees formulas like $x = A^{-1}b$, but numerically it is usually better to solve the system directly using `np.linalg.solve` rather than computing the inverse explicitly.

6.7 Aggregation and the `axis` parameter

NumPy provides functions for aggregating arrays. When applied to 2d arrays, the `axis` parameter controls the direction.

- `np.sum(x)`, `np.mean(x)`, `np.var(x)`, `np.std(x)`: sum, mean, variance, standard deviation.
- `np.min(x)`, `np.max(x)`: minimum, maximum.
- `np.cumsum(x)`: cumulative sum.
- `axis=0`: aggregate down the rows (result: one entry per column).
- `axis=1`: aggregate across the columns (result: one entry per row).

```
data = np.array([
    [1.0, 2.0, 3.0],
    [2.0, 3.0, 4.0],
    [3.0, 4.0, 5.0]
])
print("sum over all =", np.sum(data))
print("column means =", np.mean(data, axis=0))
print("row means =", np.mean(data, axis=1))
```

```
sum over all = 27.0
column means = [2. 3. 4.]
row means = [2. 3. 4.]
```

```
# standardization: subtract column mean, divide by column std
X = np.array([[1.0, 10.0], [2.0, 20.0], [3.0, 30.0], [4.0, 40.0]])
mu = np.mean(X, axis=0)
sigma = np.std(X, axis=0)
```

```
X_std = (X - mu) / sigma
print("column means after standardization =", np.mean(X_std, axis=0))
print("column stds after standardization =", np.std(X_std, axis=0))
```

```
column means after standardization = [0. 0.]
column stds after standardization = [1. 1.]
```

Standardization is a common preprocessing step. It ensures that all features are on comparable scales, which is important e.g. for gradient-based optimization.

6.8 Random numbers (`np.random`)

Randomness enters data analysis in many places: synthetic data generation, random initialization of parameters, train-test splitting, and randomized algorithms. The recommended interface is `np.random.default_rng`, which creates a random number generator with a given seed.

- `rng = np.random.default_rng(seed)`: create a random number generator.
- `rng.random(n)`: `n` uniform random numbers in $[0, 1)$.
- `rng.standard_normal(n)`: `n` standard normal random numbers.
- `rng.normal(loc, scale, size)`: normal with given mean and standard deviation.
- `rng.integers(low, high, size)`: random integers in `[low, high)`.
- `rng.choice(a, size, replace)`: random sample from array `a`.
- `rng.permutation(n)`: a random permutation of `0, ..., n-1`.

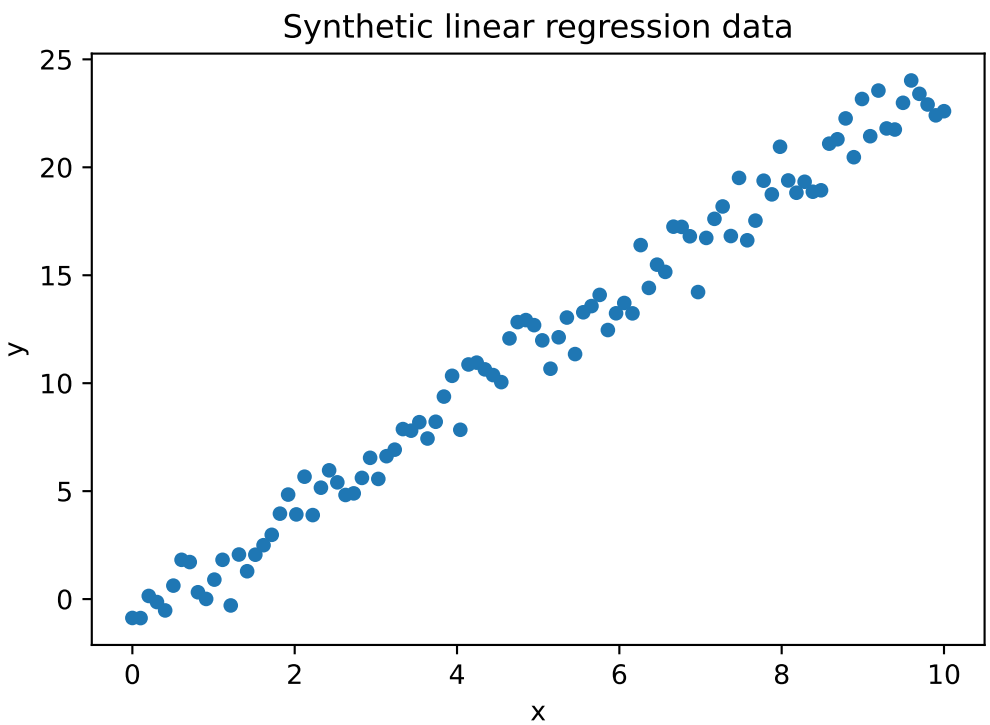
```
rng = np.random.default_rng(42)
print("uniform =", rng.random(5))
print("normal =", rng.standard_normal(5))
print("integers =", rng.integers(0, 10, size=5))
```

```
uniform = [0.77395605 0.43887844 0.85859792 0.69736803 0.09417735]
normal = [-1.30217951  0.1278404  -0.31624259 -0.01680116 -0.85304393]
integers = [5 3 1 9 7]
```

```
# reproducibility: same seed gives same numbers
rng1 = np.random.default_rng(123)
rng2 = np.random.default_rng(123)
print(rng1.standard_normal(4))
print(rng2.standard_normal(4))
```

```
[-0.98912135 -0.36778665  1.28792526  0.19397442]
[-0.98912135 -0.36778665  1.28792526  0.19397442]
```

```python
# a synthetic dataset
import matplotlib.pyplot as plt

rng = np.random.default_rng(0)
n = 100
x = np.linspace(0.0, 10.0, n)
noise = rng.normal(0.0, 1.0, size=n)
y = 2.5 * x - 1.0 + noise

plt.figure(figsize=(6, 4))
plt.scatter(x, y, s=20)
plt.xlabel("x")
plt.ylabel("y")
plt.title("Synthetic linear regression data")
plt.show()
```



```
# train-test splitting by permutation
rng = np.random.default_rng(7)
n = 100
X = rng.normal(size=(n, 3))
y = rng.normal(size=n)

perm = rng.permutation(n)
train_size = int(0.8 * n)
X_train = X[perm[:train_size]]
X_test = X[perm[train_size:]]
print("X_train.shape =", X_train.shape)
print("X_test.shape =", X_test.shape)
```

```
X_train.shape = (80, 3)
X_test.shape = (20, 3)
```

6.9 Optimization (`scipy.optimize`)

The `scipy.optimize` module provides tools for finding minima of functions and roots of equations. This is essential in many areas of mathematics and data analysis.

- `optimize.minimize_scalar(f)`: find the minimum of a scalar function $f : \mathbb{R} \to \mathbb{R}$.
- `optimize.minimize(f, x0)`: find the minimum of $f : \mathbb{R}^d \to \mathbb{R}$, starting from `x0`. Returns a result object with attributes `.x` (minimizer), `.fun` (minimum value), `.success`, `.nit` (number of iterations).
- `optimize.minimize(f, x0, jac=grad_f)`: same, but with analytically supplied gradient for faster convergence.
- `optimize.minimize(f, x0, method='...')`: choose a specific algorithm, e.g. `'BFGS'`, `'Nelder-Mead'`, `'L-BFGS-B'`.
- `optimize.root_scalar(f, bracket=[a, b])`: find a root of f in the interval $[a, b]$.

```
from scipy import optimize
```

```
# minimizing a function of one variable
def f(x):
    return (x - 3)**2 + 1

result = optimize.minimize_scalar(f)
print("minimum at x =", result.x)
print("minimum value =", result.fun)
```

```
minimum at x = 3.0
minimum value = 1.0
```

```python
# minimizing a function of several variables (Rosenbrock function)
def rosenbrock(x):
    return (1 - x[0])**2 + 100 * (x[1] - x[0]**2)**2

x0 = np.array([0.0, 0.0])
result = optimize.minimize(rosenbrock, x0)
print("minimum at x =", result.x)
print("success =", result.success)
```

```
minimum at x = [0.99999467 0.99998932]
success = True
```

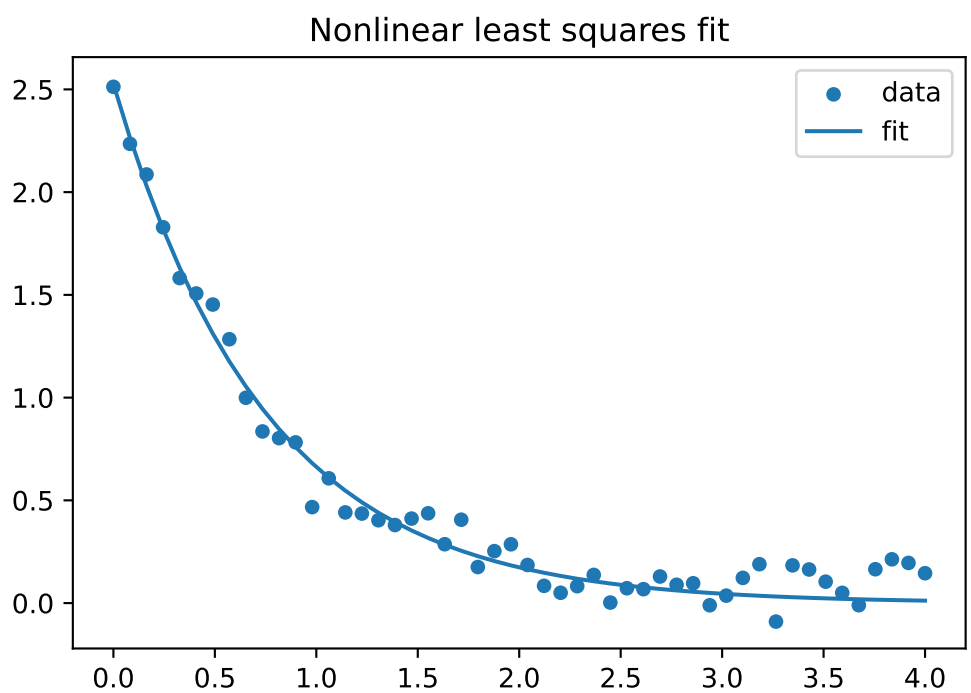
```python
# providing the gradient for faster convergence
def rosenbrock_grad(x):
    dfdx0 = -2 * (1 - x[0]) - 400 * x[0] * (x[1] - x[0]**2)
    dfdx1 = 200 * (x[1] - x[0]**2)
    return np.array([dfdx0, dfdx1])

result = optimize.minimize(rosenbrock, x0, jac=rosenbrock_grad, method='BFGS')
print("minimum at x =", result.x)
print("iterations =", result.nit)
```

```
minimum at x = [0.99999913 0.99999825]
iterations = 19
```

```python
# nonlinear least squares fitting
rng = np.random.default_rng(0)
x_data = np.linspace(0, 4, 50)
y_data = 2.5 * np.exp(-1.3 * x_data) + 0.1 * rng.standard_normal(50)

def loss(params):
    a, b = params
    return np.mean((y_data - a * np.exp(-b * x_data))**2)

result = optimize.minimize(loss, x0=[1.0, 1.0])
a_fit, b_fit = result.x
print(f"fitted parameters: a = {a_fit:.3f}, b = {b_fit:.3f}")
```

```
plt.figure(figsize=(6, 4))
plt.scatter(x_data, y_data, s=20, label="data")
plt.plot(x_data, a_fit * np.exp(-b_fit * x_data), label="fit")
plt.legend()
plt.title("Nonlinear least squares fit")
plt.show()
```

```
fitted parameters: a = 2.526, b = 1.338
```




```
# root finding
def g(x):
    return x**3 - 2*x - 5

result = optimize.root_scalar(g, bracket=[2, 3])
print("root at x =", result.root)
print("g(root) =", g(result.root))
```

```
root at x = 2.094551481542327
g(root) = 3.552713678800501e-15
```

7 Loading and handling data

Download notebook.

In this chapter we learn how to load data from various sources and how to handle tabular data using `pandas`.

7.1 Loading data from files (`csv`, `xlsx`, `json`)

Data often comes as files. The most common formats are CSV (comma-separated values), Excel (`.xlsx`), and JSON (JavaScript Object Notation). Python has built-in support for CSV and JSON, and `pandas` can read all three. The built-in modules are:

- `csv.reader(file)`: reads a CSV file row by row (each row is a list of strings).
- `csv.DictReader(file)`: reads a CSV file row by row (each row is a dictionary with column names as keys).
- `json.loads(s)`: parses a JSON string into a Python object (dict, list, etc.).
- `json.dumps(obj)`: converts a Python object to a JSON string.
- `json.load(file)` / `json.dump(obj, file)`: read/write JSON from/to a file.

```
import csv
import io

# We simulate a CSV file using a string
csv_string = """name,age,city
Alice,30,Berlin
Bob,25,Munich
Charlie,35,Hamburg"""

reader = csv.DictReader(io.StringIO(csv_string))
for row in reader:
    print(row)
```

```
{'name': 'Alice', 'age': '30', 'city': 'Berlin'}
{'name': 'Bob', 'age': '25', 'city': 'Munich'}
{'name': 'Charlie', 'age': '35', 'city': 'Hamburg'}
```

```python
import json

# parsing a JSON string
json_string = '{"name": "Alice", "age": 30, "scores": [85, 92, 78]}'
data = json.loads(json_string)
print(data)
print("name:", data["name"])
```

```
{'name': 'Alice', 'age': 30, 'scores': [85, 92, 78]}
name: Alice
```

```python
# converting back to JSON
person = {"name": "Bob", "age": 25, "city": "Munich"}
print(json.dumps(person, indent=2))
```

```
{
  "name": "Bob",
  "age": 25,
  "city": "Munich"
}
```

In practice, you will usually use `pandas` to read data files (see below). For Excel files, `pd.read_excel("data.xlsx")` requires the `openpyxl` package.

7.2 Loading data from online resources (`requests`)

The `requests` library allows downloading data from the internet.

- `requests.get(url)`: sends a GET request and returns a response object.
- `response.status_code`: HTTP status code (200 means success).
- `response.text`: the response body as a string.
- `response.json()`: parse the response body as JSON.

```python
import requests

url = "https://jsonplaceholder.typicode.com/todos/1"
response = requests.get(url)
print("status code:", response.status_code)
print("content:", response.json())
```

```
status code: 200
content: {'userId': 1, 'id': 1, 'title': 'delectus aut autem', 'completed': False}
```

You can also combine `requests` with `pandas`: `pd.read_csv("https://example.com/data.csv")` reads a CSV file directly from a URL.

7.3 Dataframes (`pandas`)

`pandas` is the standard library for handling tabular data in Python. Its central data structure is the `DataFrame`, which can be thought of as a matrix together with row and column labels.

```python
import pandas as pd
import numpy as np
```

Here, `df` is a `DataFrame`:

- `pd.DataFrame(dict)`: create a DataFrame from a dictionary (keys become column names).
- `df["col"]`: select a single column (returns a `Series`).
- `df[["col1", "col2"]]`: select multiple columns.
- `df.iloc[i]`: select row `i` by position.
- `df.loc[i, "col"]`: select by label.
- `df[df["col"] > value]`: filter rows by a condition.
- `df.head(n)` / `df.tail(n)`: first / last `n` rows.
- `df.describe()`: summary statistics for all numeric columns.
- `df.dtypes`: data types of all columns.
- `df.info()`: concise summary of the DataFrame.
- `df.sort_values("col")`: sort by a column.
- `df.groupby("col").mean()`: group by a column and compute means.
- `df.isna()`: boolean mask of missing values.
- `df.dropna()`: drop rows with missing values.
- `df.fillna(value)`: fill missing values.
- `df.drop(columns=["col"])`: remove a column.
- `df.to_numpy()`: convert to a NumPy array.
- `pd.read_csv(path)`: read a CSV file into a DataFrame.
- `pd.to_datetime(...)`: convert strings to datetime objects.

```python
df = pd.DataFrame({
    "age": [22, 27, 31, 45, 38, 29],
    "income": [42000, 51000, 64000, 83000, 72000, 56000],
    "hours_studied": [10, 12, 7, 4, 6, 9],
```

```
    "passed": [1, 1, 0, 0, 1, 1]
})
df
```

	age	income	hours_studied	passed
0	22	42000	10	1
1	27	51000	12	1
2	31	64000	7	0
3	45	83000	4	0
4	38	72000	6	1
5	29	56000	9	1

```
# selecting columns
print(df["age"])
```

```
0    22
1    27
2    31
3    45
4    38
5    29
Name: age, dtype: int64
```

```
print(df[["age", "income"]])
```

```
   age  income
0   22   42000
1   27   51000
2   31   64000
3   45   83000
4   38   72000
5   29   56000
```

```
# selecting rows
print(df.iloc[0])
```

```
age                 22
income           42000
```

```
hours_studied       10
passed               1
Name: 0, dtype: int64
```

```
print(df.loc[0:2, ["age", "passed"]])
```

```
   age  passed
0   22       1
1   27       1
2   31       0
```

```
# filtering
filtered = df[df["income"] > 55000]
filtered
```

	age	income	hours_studied	passed
2	31	64000	7	0
3	45	83000	4	0
4	38	72000	6	1
5	29	56000	9	1

```
# summary statistics
print(df.describe())
```

```
             age        income  hours_studied    passed
count   6.000000      6.000000       6.000000  6.000000
mean   32.000000  61333.333333       8.000000  0.666667
std     8.246211  14827.901627       2.898275  0.516398
min    22.000000  42000.000000       4.000000  0.000000
25%    27.500000  52250.000000       6.250000  0.250000
50%    30.000000  60000.000000       8.000000  1.000000
75%    36.250000  70000.000000       9.750000  1.000000
max    45.000000  83000.000000      12.000000  1.000000
```

```
# sorting
df_sorted = df.sort_values("income", ascending=False)
print(df_sorted)
```

```
   age  income  hours_studied  passed
3   45   83000              4       0
4   38   72000              6       1
2   31   64000              7       0
5   29   56000              9       1
1   27   51000             12       1
0   22   42000             10       1
```

```python
# groupby
grouped = df.groupby("passed").mean(numeric_only=True)
print(grouped)
```

```
         age   income  hours_studied
passed
0       38.0  73500.0           5.50
1       29.0  55250.0           9.25
```

```python
# missing values
df_missing = df.copy()
df_missing.loc[2, "income"] = np.nan
df_missing.loc[4, "hours_studied"] = np.nan

print(df_missing)
print()
print(df_missing.isna().sum())
print()
print(df_missing.dropna())
```

```
   age   income  hours_studied  passed
0   22  42000.0           10.0       1
1   27  51000.0           12.0       1
2   31      NaN            7.0       0
3   45  83000.0            4.0       0
4   38  72000.0            NaN       1
5   29  56000.0            9.0       1

age              0
income           1
hours_studied    1
passed           0
dtype: int64
```

```
   age   income  hours_studied  passed
0   22  42000.0           10.0       1
1   27  51000.0           12.0       1
3   45  83000.0            4.0       0
5   29  56000.0            9.0       1
```

```python
# fill missing values with column means
df_filled = df_missing.fillna(df_missing.mean(numeric_only=True))
print(df_filled)
```

```
   age   income  hours_studied  passed
0   22  42000.0           10.0       1
1   27  51000.0           12.0       1
2   31  60800.0            7.0       0
3   45  83000.0            4.0       0
4   38  72000.0            8.4       1
5   29  56000.0            9.0       1
```

```python
# from pandas to numpy
clean = df_missing.dropna()
X = clean[["age", "income", "hours_studied"]].to_numpy(dtype=float)
y = clean["passed"].to_numpy(dtype=float)
print("X.shape =", X.shape)
print("y =", y)
```

```
X.shape = (4, 3)
y = [1. 1. 0. 1.]
```

The guiding principle is: **pandas handles labeled tabular data; numpy performs the mathematical core.**

```python
# reading CSV files (here with a fallback if the file does not exist)
from pathlib import Path

csv_path = Path("ml_dataset.csv")

if csv_path.exists():
    df_csv = pd.read_csv(csv_path)
else:
```

```
    rng = np.random.default_rng(42)
    n = 120
    df_csv = pd.DataFrame({
        "age": rng.integers(18, 65, size=n),
        "income": rng.normal(55000, 14000, size=n),
        "hours_studied": rng.normal(8, 2, size=n),
    })
    logits = (
        -4
        + 0.00005 * df_csv["income"]
        + 0.25 * df_csv["hours_studied"]
        - 0.02 * df_csv["age"]
    )
    probs = 1 / (1 + np.exp(-logits))
    df_csv["target"] = (rng.random(n) < probs).astype(int)

df_csv.head()
```

	age	income	hours_studied	target
0	22	42878.364380	7.762723	0
1	54	68555.896964	8.571652	1
2	48	31439.823197	10.612003	1
3	38	50311.609580	8.438765	1
4	38	57278.542911	7.178146	1

```
print(df_csv.describe())
```

```
             age        income  hours_studied      target
count  120.00000    120.000000     120.000000  120.000000
mean    41.95000  53343.380771       7.991499    0.458333
std     12.83499  12735.834218       2.016080    0.500350
min     18.00000  25151.352070       2.866683    0.000000
25%     31.00000  43345.030963       6.670809    0.000000
50%     42.00000  52550.425867       7.808451    0.000000
75%     54.00000  61287.290680       8.879359    1.000000
max     63.00000  95794.074524      13.810134    1.000000
```

7.3.1 Working with dates and times

`pandas` can parse and work with dates.

- `pd.to_datetime(...)`: convert strings to datetime.
- `dt.strftime(format)`: format a datetime as a string.

```
dates = pd.to_datetime(["2024-01-15", "2024-03-22", "2024-07-01"])
print(dates)
print("Month:", dates.month.tolist())
print("Day of week:", dates.day_of_week.tolist())
```

```
DatetimeIndex(['2024-01-15', '2024-03-22', '2024-07-01'], dtype='datetime64[us]', freq=None)
Month: [1, 3, 7]
Day of week: [0, 4, 0]
```

```
dt = pd.Timestamp("2024-06-15 14:30:00")
print(dt.strftime("%-m/%-d/%y, %H:%M"))
```

```
6/15/24, 14:30
```

7.4 SQL databases (`sqlite3`)

Python has built-in support for SQLite databases via the `sqlite3` module. This is useful for working with structured data that lives in a database.

- `sqlite3.connect(path)`: connect to a database (use `":memory:"` for an in-memory database).
- `cursor.execute(sql)`: execute an SQL statement.
- `cursor.fetchall()`: fetch all results.
- `pd.read_sql_query(sql, conn)`: run an SQL query and return the result as a DataFrame.

```
import sqlite3

conn = sqlite3.connect(":memory:")
cursor = conn.cursor()

cursor.execute("""
    CREATE TABLE students (
        name TEXT,
        age INTEGER,
        grade REAL
    )
```

```python
""")

students = [
    ("Alice", 22, 1.3),
    ("Bob", 25, 2.0),
    ("Charlie", 23, 1.7),
    ("Diana", 27, 2.3),
]

cursor.executemany("INSERT INTO students VALUES (?, ?, ?)", students)
conn.commit()
```

```python
# querying
cursor.execute("SELECT * FROM students WHERE age > 23")
for row in cursor.fetchall():
    print(row)
```

```
('Bob', 25, 2.0)
('Diana', 27, 2.3)
```

```python
# reading SQL results into a pandas DataFrame
df_sql = pd.read_sql_query("SELECT * FROM students", conn)
print(df_sql)
conn.close()
```

```
      name  age  grade
0    Alice   22    1.3
1      Bob   25    2.0
2  Charlie   23    1.7
3    Diana   27    2.3
```

8 Analyzing data

Download notebook.

In this chapter, we learn about visualization with `matplotlib` and statistical analysis with `scipy.stats`, including maximum likelihood estimation.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

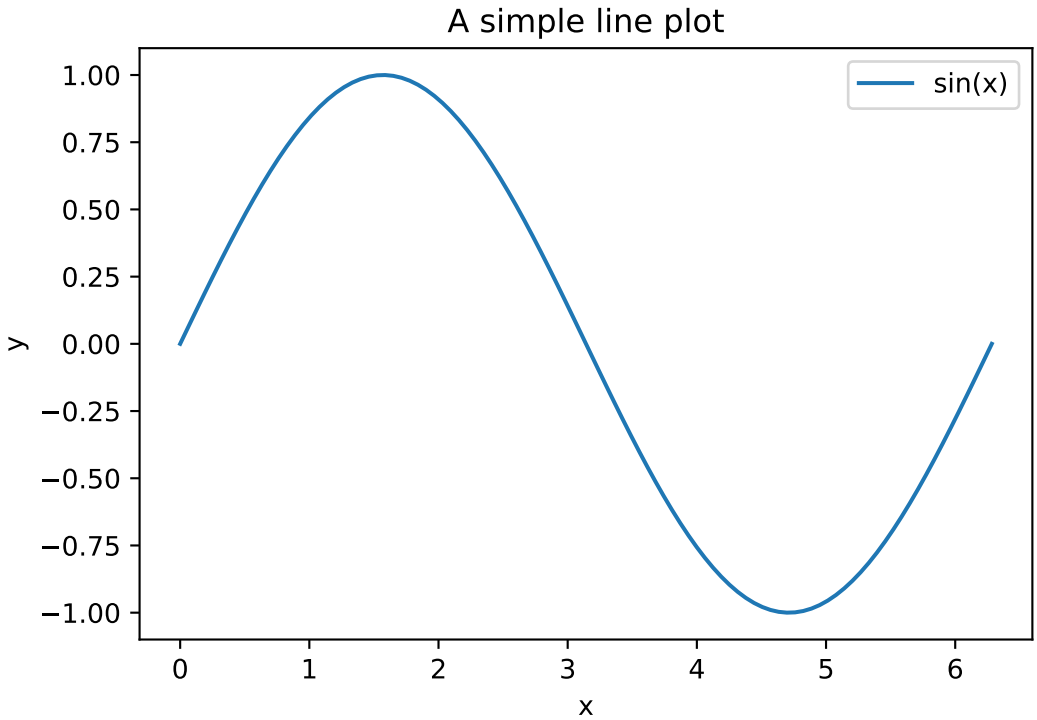
8.1 Visualization (`matplotlib`)

The standard plotting library in Python is `matplotlib`. Its `pyplot` module provides a MATLAB-like interface. Here, `plt` refers to `matplotlib.pyplot`.

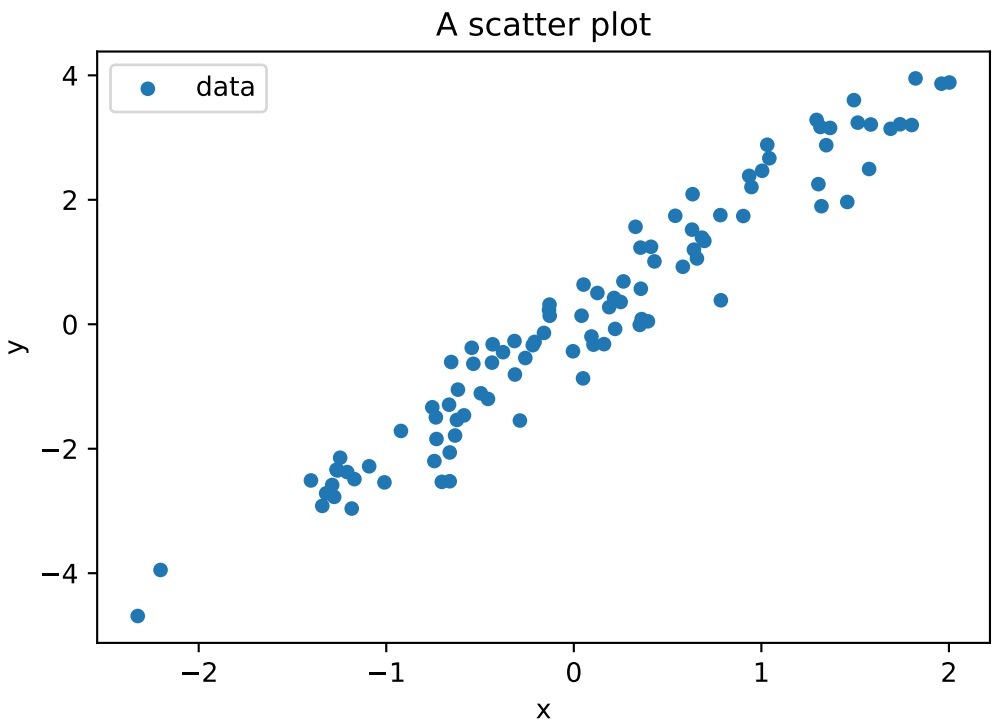
- `plt.figure(figsize=(w, h))`: create a new figure with given size in inches.
- `plt.plot(x, y, label=...)`: line plot.
- `plt.scatter(x, y, s=..., label=...)`: scatter plot (`s` is marker size).
- `plt.hist(data, bins=..., density=...)`: histogram.
- `plt.bar(categories, values)`: bar plot.
- `plt.boxplot(data_list)`: box plot.
- `plt.xlabel(...)`, `plt.ylabel(...)`, `plt.title(...)`: axis labels and title.
- `plt.legend()`: show legend.
- `plt.show()`: display the figure.
- `fig, axes = plt.subplots(nrows, ncols)`: create a figure with multiple subplots.

```
x = np.linspace(0, 2 * np.pi, 100)
y = np.sin(x)

plt.figure(figsize=(6, 4))
plt.plot(x, y, label="sin(x)")
plt.xlabel("x")
plt.ylabel("y")
plt.title("A simple line plot")
```

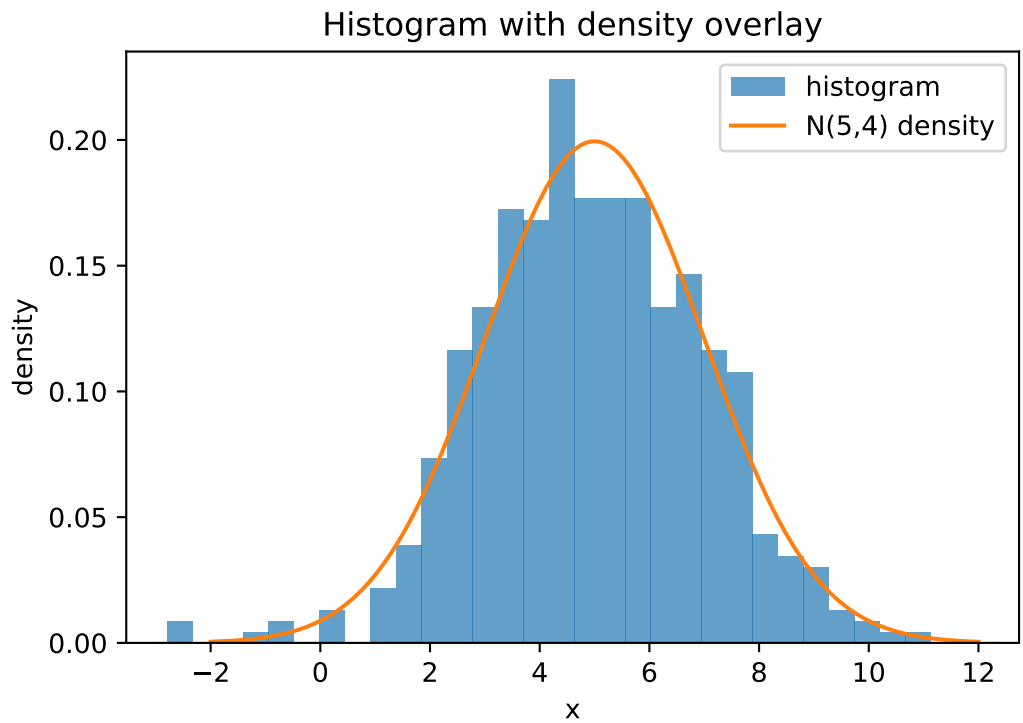
```
plt.legend()
plt.show()
```



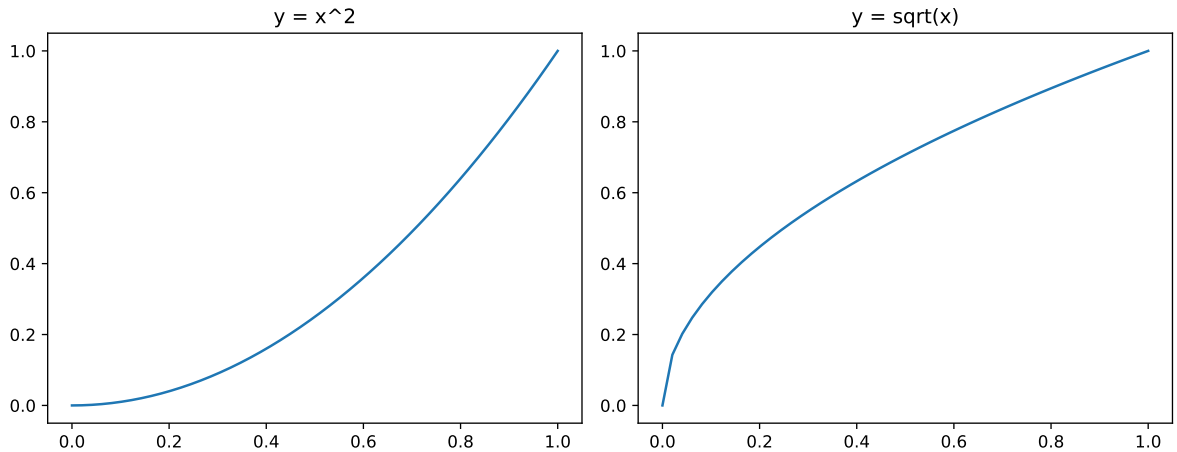

```
rng = np.random.default_rng(0)
x = rng.normal(size=100)
y = 2 * x + 0.5 * rng.normal(size=100)

plt.figure(figsize=(6, 4))
plt.scatter(x, y, s=20, label="data")
plt.xlabel("x")
plt.ylabel("y")
plt.title("A scatter plot")
plt.legend()
plt.show()
```




```
# histogram with density overlay
data = rng.normal(loc=5, scale=2, size=500)
x_grid = np.linspace(-2, 12, 200)

plt.figure(figsize=(6, 4))
plt.hist(data, bins=30, density=True, alpha=0.7, label="histogram")
plt.plot(x_grid, stats.norm.pdf(x_grid, loc=5, scale=2), label="N(5,4) density")
plt.xlabel("x")
plt.ylabel("density")
plt.title("Histogram with density overlay")
plt.legend()
plt.show()
```

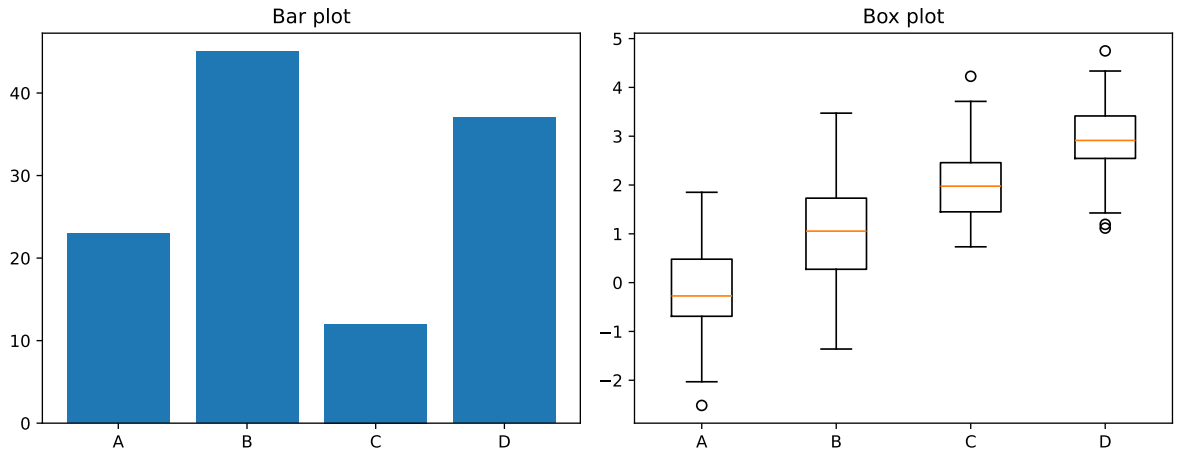



```
# subplots
fig, axes = plt.subplots(1, 2, figsize=(10, 4))

axes[0].plot(np.linspace(0, 1, 50), np.linspace(0, 1, 50)**2)
axes[0].set_title("y = x^2")

axes[1].plot(np.linspace(0, 1, 50), np.sqrt(np.linspace(0, 1, 50)))
axes[1].set_title("y = sqrt(x)")

plt.tight_layout()
plt.show()
```




```
# bar plot and box plot
categories = ["A", "B", "C", "D"]
values = [23, 45, 12, 37]

fig, axes = plt.subplots(1, 2, figsize=(10, 4))

axes[0].bar(categories, values)
axes[0].set_title("Bar plot")

data_groups = [rng.normal(loc=m, size=50) for m in [0, 1, 2, 3]]
axes[1].boxplot(data_groups, labels=categories)
axes[1].set_title("Box plot")

plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_30236/117098709.py:11: MatplotlibDeprecationWarning: The 'labels' parameter o
  axes[1].boxplot(data_groups, labels=categories)
```



8.2 Descriptive statistics (`scipy.stats`)

The `scipy.stats` module provides functions for descriptive statistics and statistical tests. Here, `data` is a 1d array or list.

- `stats.describe(data)`: returns count, min, max, mean, variance, skewness, kurtosis.
- `stats.tmean(data)`: trimmed mean.
- `stats.tstd(data)`: trimmed standard deviation.
- `stats.tvar(data)`: trimmed variance.
- `stats.pearsonr(x, y)`: Pearson correlation coefficient and p-value.
- `stats.spearmanr(x, y)`: Spearman rank correlation and p-value.

```
data = [1, 2, 3, 4, 5, 6, 7, 8]
result = stats.describe(data)
print(result)
```

```
DescribeResult(nobs=np.int64(8), minmax=(np.int64(1), np.int64(8)), mean=np.float64(4.5), va
1.2380952380952381))
```

```
data = np.array([2.3, 4.1, 3.7, 5.2, 4.8, 3.3])
print("trimmed mean =", stats.tmean(data))
print("std =", stats.tstd(data))
print("median =", np.median(data))
```

```
trimmed mean = 3.9000000000000004
std = 1.0488088481701514
median = 3.9
```

```
# correlation
rng = np.random.default_rng(0)
x = rng.normal(size=50)
y = 2 * x + 0.5 * rng.normal(size=50)

r, p_value = stats.pearsonr(x, y)
print(f"Pearson r = {r:.4f}, p-value = {p_value:.6f}")
```

```
Pearson r = 0.9621, p-value = 0.000000
```

8.3 Probability distributions (`scipy.stats`)

`scipy.stats` provides a large number of probability distributions. Each distribution object has a common interface. Here, `dist` is a distribution object (e.g. `stats.norm`, `stats.expon`, `stats.poisson`).

- `dist.rvs(size=n)`: draw `n` random samples.
- `dist.pdf(x)` / `dist.pmf(x)`: density / probability mass function.
- `dist.cdf(x)`: cumulative distribution function.
- `dist.ppf(q)`: quantile function (inverse of `cdf`).
- `dist.mean()`, `dist.var()`, `dist.std()`: theoretical moments.
- `dist.fit(data)`: estimate parameters from data (maximum likelihood).

For continuous distributions, the location-scale convention is used: `stats.norm(loc=mu, scale=sigma)` gives $\mathcal{N}(\mu, \sigma^2)$.

```
# standard normal
print("mean =", stats.norm.mean())
print("pdf(0) =", stats.norm.pdf(0))
print("cdf(1.96) =", stats.norm.cdf(1.96))
print("ppf(0.975) =", stats.norm.ppf(0.975))
```

```
mean = 0.0
pdf(0) = 0.3989422804014327
cdf(1.96) = 0.9750021048517795
ppf(0.975) = 1.959963984540054
```

```
# normal with given parameters
dist = stats.norm(loc=3, scale=2)
samples = dist.rvs(size=5, random_state=42)
print("samples:", samples)
print("mean =", dist.mean())
print("variance =", dist.var())
```

```
samples: [3.99342831 2.7234714  4.29537708 6.04605971 2.53169325]
mean = 3.0
variance = 4.0
```

```
# exponential
print("Exp(1) mean =", stats.expon.mean())
print("Exp(1) samples:", stats.expon.rvs(size=5, random_state=0))
```

```
Exp(1) mean = 1.0
Exp(1) samples: [0.79587451 1.25593076 0.92322315 0.78720115 0.55104849]
```

```
# Poisson
print("Poisson(3) pmf(2) =", stats.poisson.pmf(2, mu=3))
```

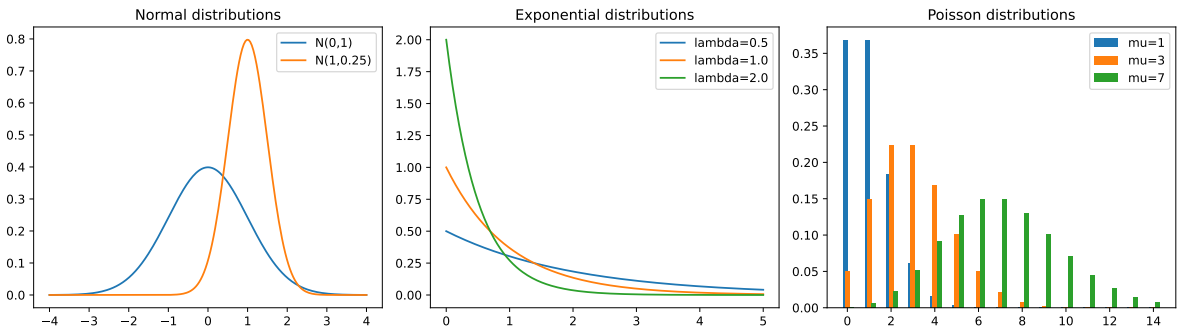
```
Poisson(3) pmf(2) = 0.22404180765538775
```

```
# binomial
print("Binom(10, 0.3) pmf(3) =", stats.binom.pmf(3, n=10, p=0.3))
print("Binom(10, 0.3) mean =", stats.binom.mean(n=10, p=0.3))
```

```
Binom(10, 0.3) pmf(3) = 0.2668279319999998
Binom(10, 0.3) mean = 3.0
```

```
# visualizing distributions
fig, axes = plt.subplots(1, 3, figsize=(14, 4))

x = np.linspace(-4, 4, 200)
axes[0].plot(x, stats.norm.pdf(x), label="N(0,1)")
axes[0].plot(x, stats.norm.pdf(x, loc=1, scale=0.5), label="N(1,0.25)")
axes[0].set_title("Normal distributions")
axes[0].legend()
```

```
x = np.linspace(0, 5, 200)
for lam in [0.5, 1.0, 2.0]:
    axes[1].plot(x, stats.expon.pdf(x, scale=1/lam), label=f"lambda={lam}")
axes[1].set_title("Exponential distributions")
axes[1].legend()

k = np.arange(0, 15)
for mu in [1, 3, 7]:
    axes[2].bar(k + 0.2*(mu-3)/4, stats.poisson.pmf(k, mu=mu), width=0.2, label=f"mu={mu}")
axes[2].set_title("Poisson distributions")
axes[2].legend()

plt.tight_layout()
plt.show()
```



8.4 Statistical tests

`scipy.stats` provides many standard statistical tests. Each test returns a test statistic and a p-value.

- `stats.ttest_1samp(data, popmean)`: one-sample t-test (is the mean equal to `popmean`?).
- `stats.ttest_ind(sample1, sample2)`: two-sample t-test (do two independent samples have the same mean?).
- `stats.chisquare(observed)`: chi-squared test (do observed frequencies match expected frequencies?).
- `stats.kstest(data, 'norm')`: Kolmogorov-Smirnov test (does the data come from the given distribution?).

```python
rng = np.random.default_rng(0)
data = rng.normal(loc=5.2, scale=1.0, size=30)

# one-sample t-test
t_stat, p_value = stats.ttest_1samp(data, popmean=5.0)
print(f"t-statistic = {t_stat:.4f}, p-value = {p_value:.4f}")
```

```
t-statistic = 0.5225, p-value = 0.6053
```

```python
# two-sample t-test
sample1 = rng.normal(loc=5.0, scale=1.0, size=40)
sample2 = rng.normal(loc=5.5, scale=1.0, size=40)

t_stat, p_value = stats.ttest_ind(sample1, sample2)
print(f"t-statistic = {t_stat:.4f}, p-value = {p_value:.4f}")
```

```
t-statistic = -1.2000, p-value = 0.2338
```

```python
# chi-squared test
observed = np.array([18, 22, 20, 25, 15])
stat, p_value = stats.chisquare(observed)
print(f"chi2-statistic = {stat:.4f}, p-value = {p_value:.4f}")
```

```
chi2-statistic = 2.9000, p-value = 0.5747
```

```python
# Kolmogorov-Smirnov test
data = rng.normal(loc=0, scale=1, size=100)
stat, p_value = stats.kstest(data, 'norm')
print(f"KS statistic = {stat:.4f}, p-value = {p_value:.4f}")
```

```
KS statistic = 0.0477, p-value = 0.9689
```

8.5 Maximum likelihood estimation

Maximum likelihood estimation (MLE) is one of the most important methods in statistics. Given i.i.d. observations $x_1, \dots, x_n$ from a distribution with density $f(x;\theta)$, the likelihood function is

$$L(\theta) = \prod_{i=1}^{n} f(x_i;\theta),$$

and the log-likelihood is

$$\ell(\theta) = \sum_{i=1}^{n} \log f(x_i; \theta).$$

The MLE is the parameter value that maximizes $\ell(\theta)$, or equivalently minimizes $-\ell(\theta)$.

In `scipy.stats`, the `fit` method computes the MLE for any distribution:

- `dist.fit(data)`: returns the MLE of the distribution parameters.
- `dist.logpdf(x, ...)`: log-density, useful for computing the log-likelihood.

For cases where no closed-form MLE exists, we use numerical optimization via `scipy.optimize.minimize` (see Section 6.9).

```
# MLE for the normal distribution
rng = np.random.default_rng(0)
true_mu = 3.0
true_sigma = 1.5
data = rng.normal(loc=true_mu, scale=true_sigma, size=200)

# analytical MLE
mu_hat = np.mean(data)
sigma_hat = np.std(data)  # np.std uses 1/n, which is the MLE
print(f"true mu = {true_mu}, MLE mu = {mu_hat:.4f}")
print(f"true sigma = {true_sigma}, MLE sigma = {sigma_hat:.4f}")
```

```
true mu = 3.0, MLE mu = 3.0229
true sigma = 1.5, MLE sigma = 1.4418
```

```
# using scipy's fit method
mu_fit, sigma_fit = stats.norm.fit(data)
print(f"scipy fit: mu = {mu_fit:.4f}, sigma = {sigma_fit:.4f}")
```

```
scipy fit: mu = 3.0229, sigma = 1.4418
```

```
# MLE for the exponential distribution
# For Exp(lambda) with density lambda * exp(-lambda * x), the MLE is lambda_hat = 1 / x_bar
true_lambda = 2.0
data_exp = rng.exponential(scale=1/true_lambda, size=150)

lambda_hat = 1 / np.mean(data_exp)
print(f"true lambda = {true_lambda}, MLE lambda = {lambda_hat:.4f}")
```

```
true lambda = 2.0, MLE lambda = 1.7986
```

```python
# MLE via numerical optimization (when no closed form exists)
# Example: Gamma distribution
from scipy import optimize

true_shape = 2.5
true_scale = 1.3
data_gamma = rng.gamma(shape=true_shape, scale=true_scale, size=300)

def neg_log_likelihood_gamma(params):
    a, scale = params
    if a <= 0 or scale <= 0:
        return np.inf
    return -np.sum(stats.gamma.logpdf(data_gamma, a=a, scale=scale))

result = optimize.minimize(neg_log_likelihood_gamma, x0=[1.0, 1.0], method='Nelder-Mead')
a_hat, scale_hat = result.x
print(f"true shape = {true_shape}, MLE shape = {a_hat:.4f}")
print(f"true scale = {true_scale}, MLE scale = {scale_hat:.4f}")
```

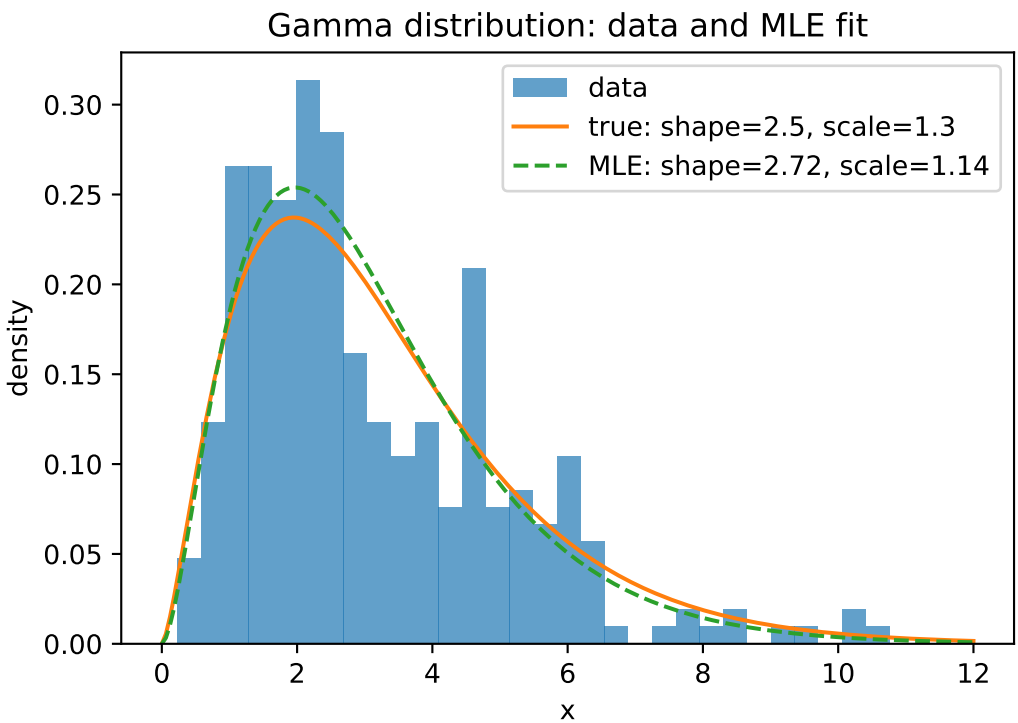
```
true shape = 2.5, MLE shape = 2.7228
true scale = 1.3, MLE scale = 1.1411
```

```python
# compare with scipy's built-in fit
a_fit, loc_fit, scale_fit = stats.gamma.fit(data_gamma, floc=0)
print(f"scipy fit: shape = {a_fit:.4f}, scale = {scale_fit:.4f}")
```

```
scipy fit: shape = 2.7227, scale = 1.1411
```

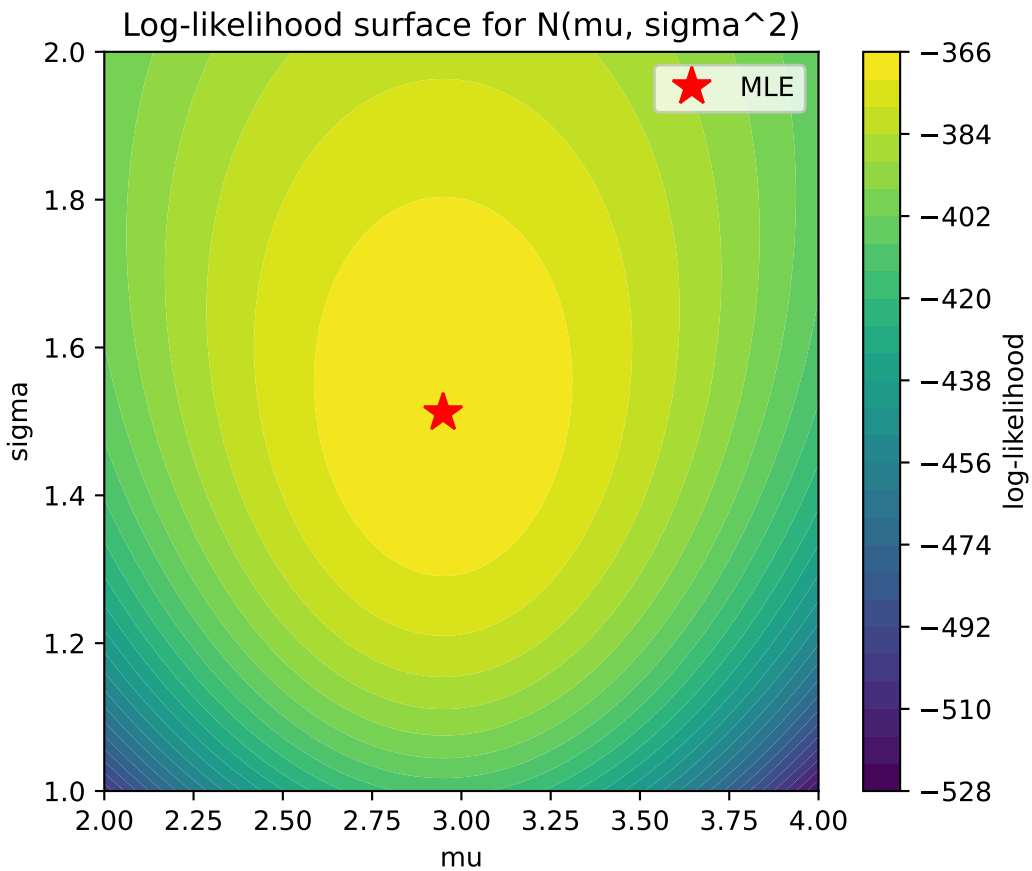
```python
# visualizing the MLE fit
x_grid = np.linspace(0, 12, 200)

plt.figure(figsize=(6, 4))
plt.hist(data_gamma, bins=30, density=True, alpha=0.7, label="data")
plt.plot(x_grid, stats.gamma.pdf(x_grid, a=true_shape, scale=true_scale),
         label=f"true: shape={true_shape}, scale={true_scale}")
plt.plot(x_grid, stats.gamma.pdf(x_grid, a=a_hat, scale=scale_hat),
         label=f"MLE: shape={a_hat:.2f}, scale={scale_hat:.2f}", linestyle="--")
plt.xlabel("x")
plt.ylabel("density")
```

```
plt.title("Gamma distribution: data and MLE fit")
plt.legend()
plt.show()
```




```
# log-likelihood surface for N(mu, sigma^2)
data = rng.normal(loc=3.0, scale=1.5, size=200)

mu_grid = np.linspace(2.0, 4.0, 100)
sigma_grid = np.linspace(1.0, 2.0, 100)
MU, SIGMA = np.meshgrid(mu_grid, sigma_grid)

log_lik = np.zeros_like(MU)
for i in range(len(sigma_grid)):
    for j in range(len(mu_grid)):
        log_lik[i, j] = np.sum(stats.norm.logpdf(data, loc=MU[i, j], scale=SIGMA[i, j]))

plt.figure(figsize=(6, 5))
plt.contourf(MU, SIGMA, log_lik, levels=30, cmap="viridis")
plt.colorbar(label="log-likelihood")
plt.xlabel("mu")
```

```
plt.ylabel("sigma")
plt.title("Log-likelihood surface for N(mu, sigma^2)")
mu_mle = np.mean(data)
sigma_mle = np.std(data)
plt.plot(mu_mle, sigma_mle, "r*", markersize=15, label="MLE")
plt.legend()
plt.show()
```



Plotting the log-likelihood surface helps build geometric intuition: the MLE is the peak of the surface. For well-behaved models, the surface is concave near the maximum, and the curvature encodes the precision of the estimate (related to the Fisher information).

9 An example project

Download notebook.

Human population data

10 Miscellaneous

We collect here some stuff, which does not really fit anywhere else. We use the material in this section without notice in the whole course.

10.1 Comments

In Python, the key for a comment is #.

```
# This line is a comment.
print("This line is not a comment.")
```

```
This line is not a comment.
```

10.2 Two equality signs

In Python (and many other programming languages), there are assignments such as `x = 0`, which says that `x` should from now on have the value `0`, and `x == 0`. The latter is a condition, which results in `True` if `x` is indeed `0`, and 'False otherwise.

10.3 User input

Sometimes, when operating on the command line, one wants to ask for user input. This works as follows:

```
name = input("What is your name?")
print(f"Hello, {name}!")
```

10.4 Regular expressions

ask the AI!

10.5 dot-notation

Here is another very useful notation convention, known as dot-notation: For a function `str.something(s,...)`, where `s` is of type `str`, you can as well write `s.something()`. For example, `s.strip()` is the same as `str.strip(s)`.

10.6 Virtual environments and `pip` (`venv`, `pip`)

claude generated

When working on different Python projects, you may need different versions of the same library. Virtual environments solve this by creating an isolated Python installation per project. The `pip` tool installs packages into the active environment.

- `python3 -m venv venv`: create a virtual environment in the folder `venv`.
- `source venv/bin/activate`: activate the virtual environment (Linux/Mac). On Windows: `venv\Scripts\activate`.
- `deactivate`: leave the virtual environment.
- `pip install package`: install a package into the active environment.
- `pip install -r requirements.txt`: install all packages listed in `requirements.txt`.
- `pip freeze`: list all installed packages and their versions.
- `pip freeze > requirements.txt`: save the current package list to a file, so others can reproduce your environment.

```
# typical workflow when starting a new project
python3 -m venv venv
source venv/bin/activate
pip install numpy pandas matplotlib scipy
pip freeze > requirements.txt
```

```
# typical workflow when cloning an existing project
git clone https://github.com/someone/some_project
cd some_project
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
```

Without a virtual environment, installing packages changes your system-wide Python installation, which can break other projects. It is good practice to always work inside a virtual environment.

10.7 Debugging (`print`, `breakpoint`)

claude generated

When code does not behave as expected, we need to find the bug. The simplest strategy is adding `print` statements to inspect variables at critical points. For more complex issues, Python has a built-in interactive debugger.

- `print(x)`: the simplest debugging tool. Print the value (and possibly the type) of a variable at a critical point.
- `print(f"{x = }")`: a shorthand (since Python 3.8) that prints both the variable name and its value.
- `type(x)`: check the type of a variable. Many bugs come from unexpected types.
- `breakpoint()`: drop into the interactive debugger (`pdb`) at this point. You can then inspect variables, step through code, and continue execution.
- In the debugger: `n` (next line), `s` (step into function), `c` (continue), `p x` (print `x`), `q` (quit).

```python
# using print for debugging
def buggy_mean(values):
    total = 0
    for v in values:
        total += v
    print(f"{total = }, {len(values) = }")  # inspect intermediate values
    return total / len(values)

result = buggy_mean([10, 20, 30])
print(f"{result = }")
```

```
total = 60, len(values) = 3
result = 20.0
```

```python
# checking types
x = "42"
print(f"{x = }, {type(x) = }")
# this would be a subtle bug: x is a string, not an int
```

```
x = '42', type(x) = <class 'str'>
```

Using `breakpoint()` is especially useful in scripts (not in notebooks). When Python hits `breakpoint()`, it opens an interactive prompt where you can inspect all variables. A typical workflow is:

```
# in your script:
def process(data):
    result = data * 2
    breakpoint()       # execution pauses here
    return result + 1
```

A general debugging strategy is:

1. Read the error message carefully (especially the last line and the line number).
2. Add `print` statements before the failing line to check variable values and types.
3. If the issue is complex, use `breakpoint()` to explore interactively.
4. Once fixed, remove the debugging statements.

10.8 Encoding and Unicode (`utf-8`, `encode`, `decode`)

claude generated

Text in a computer is stored as a sequence of bytes. An **encoding** defines how characters are mapped to bytes. The most common encoding today is `utf-8`, which can represent all Unicode characters (including ä, ö, ü, é, 、). Problems arise when a file was written in one encoding but read in another.

- `s.encode("utf-8")`: convert a string to bytes using the given encoding.
- `b.decode("utf-8")`: convert bytes back to a string.
- `open(path, "r", encoding="utf-8")`: read a file with a specific encoding. If omitted, Python uses the system default (usually `utf-8` on modern systems, but not always).
- `"latin-1"` (also called `"iso-8859-1"`): an older encoding common in Western European data. It can represent ä, ö, ü but not e.g. Chinese characters.

```python
s = "Üniversität Freiburg"
b = s.encode("utf-8")
print(f"string: {s}")
print(f"bytes:  {b}")
print(f"length of string: {len(s)}")
print(f"length of bytes:  {len(b)}")
```

```
string: Üniversität Freiburg
bytes:  b'\xc3\x9cniversit\xc3\xa4t Freiburg'
length of string: 20
length of bytes:  22
```

```
# decoding bytes back to a string
print(b.decode("utf-8"))
```

```
Üniversität Freiburg
```

```
# what goes wrong with the wrong encoding
b_latin = s.encode("latin-1")
print(f"latin-1 bytes: {b_latin}")
# decoding latin-1 bytes as utf-8 would fail:
try:
    print(b_latin.decode("utf-8"))
except UnicodeDecodeError as e:
    print(f"Error: {e}")
```

```
latin-1 bytes: b'\xdcniversit\xe4t Freiburg'
Error: 'utf-8' codec can't decode byte 0xdc in position 0: invalid continuation byte
```

```
# reading a file with explicit encoding
with open("example_utf8.txt", "w", encoding="utf-8") as f:
    f.write("Ä, Ö, Ü, ß\n")

with open("example_utf8.txt", "r", encoding="utf-8") as f:
    print(f.read())
```

```
Ä, Ö, Ü, ß
```

When you encounter a `UnicodeDecodeError` while reading a file, the most common fix is to specify the correct encoding in `open(..., encoding="...")`. Common candidates are `"utf-8"`, `"latin-1"`, and `"cp1252"` (Windows). If you are unsure about the encoding of a file, the `chardet` library can help guess it.

10.9 git and github

`git` is a version control system: it tracks changes to files over time, so you can go back to earlier versions, collaborate with others, and see who changed what and when. `github` is a website that hosts `git` repositories online.

These course notes are organized using `git`. You need to install `git` on your system. Here are the most important `git` commands:

Getting started:

- `git clone url`: download a repository from github to your computer.
- `git pull`: update your local copy with the latest changes from github.
- `git status`: show which files have been changed, added, or deleted since the last commit.
- `git log`: show the history of commits (press `q` to quit).
- `git log --oneline`: same, but one line per commit.

Making changes (for your own projects):

- `git init`: turn the current folder into a new git repository.
- `git add file`: stage a file for the next commit (i.e. mark it to be included).
- `git add .`: stage all changed files.
- `git commit -m "message"`: save the staged changes as a new commit with a description.
- `git diff`: show what has changed since the last commit.
- `git diff --staged`: show what has been staged for the next commit.

Working with github:

- `git push`: upload your commits to github.
- `git remote -v`: show which github repository your local copy is connected to.

Undoing things:

- `git checkout -- file`: discard local changes to a file (revert to last commit).
- `git reset HEAD file`: unstage a file (undo `git add`).

A typical workflow looks like this:

```
# 1. make some changes to files
# 2. check what changed
git status
git diff
# 3. stage and commit
git add file1.py file2.py
git commit -m "add data loading function"
# 4. upload to github
git push
```

The `.gitignore` file lists files and folders that git should ignore (e.g. `venv/`, `__pycache__/`, `.ipynb_checkpoints/`, data files). This prevents large or private files from being tracked.

For this course, you only need `clone` and `pull`:

```
git clone https://github.com/pfaffelh/python_for_data
cd python_for_data
```

The course notes will be updated during the semester. To get the latest version, run `git pull`. There is one catch: if you have changed a course file locally, `git pull` will refuse to overwrite your changes. The best way to avoid this is to do all your work in the folder `myFiles` or only change the `.ipynb` files — these are excluded from updates via `.gitignore`.

We recommend the following setup:

1. Clone the repository using `git clone https://github.com/pfaffelh/python_for_data` and `cd python_for_data`.
2. Create a virtual environment using `python3 -m venv venv` (see Section 10.6).
3. Activate it by `source venv/bin/activate`.
4. Install the libraries using `pip install -r requirements.txt`.

11 Additional exercises

Download notebook.

Here are additional exercises, organized by topic. They range from easy to hard within each section.

11.1 Simple data types (`int`, `float`, `str`, `bool`)

Exercise 1: Write a function `is_palindrome(s)` that returns `True` if the string `s` reads the same forwards and backwards (e.g. `"racecar"`), and `False` otherwise. Ignore upper/lower case.

Exercise 2: Given a float `x`, write a one-liner that rounds `x` to 3 decimal places without using `round`. (Hint: use arithmetic with `int()`.)

Exercise 3: Write a function `caesar(s, k)` that shifts every letter in the string `s` by `k` positions in the alphabet (wrapping around from `z` to `a`). Non-letter characters should remain unchanged. (Hint: use `ord()` and `chr()`.)

Exercise 4: IEEE 754 floating point numbers satisfy `0.1 + 0.2 != 0.3`. Write a function `approx_equal(x, y, tol=1e-9)` that returns `True` if `|x - y| < tol`. Then use it to verify that `0.1 + 0.2` is approximately `0.3`. Discuss: is a relative or absolute tolerance more appropriate in general?

11.2 Control flow (`if`, `for`, `while`, `def`)

Exercise 1: Write a function `collatz(n)` that returns the number of steps it takes for the Collatz sequence starting at `n` to reach `1`. (The rule is: if `n` is even, divide by 2; if odd, compute `3n+1`.)

Exercise 2: Write a function `primes_up_to(n)` that returns a list of all prime numbers up to `n`, using the Sieve of Eratosthenes.

Exercise 3: Write a function `newton_sqrt(a, tol=1e-10)` that computes $\sqrt{a}$ using Newton's method, i.e. the iteration $x_{k+1} = \frac{1}{2}\left(x_k + \frac{a}{x_k}\right)$, starting from $x_0 = a$, and stopping when $|x_{k+1} - x_k| < \text{tol}$.

Exercise 4: Write a function `bisection(f, a, b, tol=1e-10)` that finds a root of a continuous function f on $[a, b]$ using the bisection method. Assume $f(a)$ and $f(b)$ have opposite signs. Test it on $f(x) = x^3 - 2$.

11.3 Container data types (`list`, `dict`, `set`, `tuple`)

Exercise 1: Given a list of integers, write a function `most_frequent(lst)` that returns the element that appears most often. If there is a tie, return any of the most frequent ones. (Hint: use a `dict` to count.)

Exercise 2: Write a function `flatten(lst)` that takes a nested list (e.g. `[[1, 2], [3, [4, 5]], 6]`) and returns a flat list `[1, 2, 3, 4, 5, 6]`. Use recursion.

Exercise 3: Write a function `invert_dict(d)` that swaps keys and values of a dictionary. If multiple keys have the same value, collect the keys in a list. For example, `{"a": 1, "b": 2, "c": 1}` becomes `{1: ["a", "c"], 2: ["b"]}`.

Exercise 4: A sparse vector can be represented as a `dict` mapping indices to nonzero values, e.g. `{0: 3.0, 5: -1.0, 99: 2.0}` for a vector that is zero everywhere except at positions 0, 5, 99. Write functions `sparse_add(v, w)` and `sparse_dot(v, w)` that compute the sum and inner product of two sparse vectors.

11.4 Libraries (`math`, `datetime`, `re`, `os`)

Exercise 1: Using only the `math` module, write a function `taylor_sin(x, n)` that computes $\sin(x)$ using the first `n` terms of the Taylor series $\sin(x) = \sum_{k=0}^{n-1} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$. Compare the result with `math.sin(x)`.

Exercise 2: Write a function `days_until_birthday(birthday_str)` that takes a date string like `"1990-03-15"` and returns the number of days until the next occurrence of that birthday. (Hint: use `datetime.date`.)

Exercise 3: Write a function `extract_emails(text)` that uses a regular expression to find all email addresses in a string. Test it on a string containing several email addresses mixed with other text.

Exercise 4: Write a function `find_large_files(directory, min_size_mb=10)` that uses `os.walk` to find all files in `directory` (and subdirectories) that are larger than `min_size_mb` megabytes. Return a list of `(path, size_mb)` tuples, sorted by size descending.

11.5 Numerics (`np.array`, `np.linalg`, `np.random`)

Exercise 1: Create a 5×5 matrix A with $A_{ij} = \frac{1}{i+j+1}$ (the Hilbert matrix). Compute its condition number. Then solve $Ax = b$ for $b = (1, 1, 1, 1, 1)^\top$ and verify the result by computing $\|Ax - b\|$.

Exercise 2: Write a function `power_method(A, num_iter=100)` that approximates the largest eigenvalue of a matrix A using the power iteration: start with a random vector v, repeatedly compute $v \leftarrow Av/\|Av\|$, and return $v^\top Av$. Compare the result with `np.linalg.eig`.

Exercise 3: Generate $n = 500$ points (x_i, y_i) from the model $y = \sin(x) + \varepsilon$ where $x \in [0, 2\pi]$ and $\varepsilon \sim \mathcal{N}(0, 0.3^2)$. Fit polynomial models of degrees 3, 5, 10, 20 using the normal equation. Plot all fits and discuss overfitting.

Exercise 4: Implement gradient descent for the function $f(x, y) = (1-x)^2 + 100(y-x^2)^2$ (Rosenbrock). Experiment with different learning rates and starting points. Plot the trajectory of (x_k, y_k) on top of a contour plot of f. Compare with the result of `scipy.optimize.minimize`.

11.6 Handling data (`pandas`, `csv`, `json`)

Exercise 1: Create a DataFrame with columns `name`, `exam1`, `exam2`, `exam3` for 10 students (you can invent the data). Add a column `average` that contains the mean of the three exams. Filter to show only students with an average above 70.

Exercise 2: Download a CSV file from the internet using `pd.read_csv(url)` (e.g. from this collection). Use `describe()`, `groupby()`, and a histogram to explore the data. Write a short summary of what you find.

Exercise 3: Create a DataFrame with some missing values (`np.nan`). Compare three strategies for handling them: (a) drop rows with any missing value, (b) fill with the column mean, (c) fill with the column median. How do the summary statistics (`describe()`) change in each case?

Exercise 4: Write a function `csv_to_sqlite(csv_path, db_path, table_name)` that reads a CSV file into a pandas DataFrame and writes it into an SQLite database. Then write a function `query_db(db_path, sql)` that runs an SQL query and returns the result as a DataFrame. Test with a CSV file of your choice.

11.7 Analyzing data (`matplotlib`, `scipy.stats`, MLE)

Exercise 1: Generate 1000 samples from a $\mathcal{N}(2, 3^2)$ distribution. Plot a histogram together with the true density, the density of the MLE fit (`stats.norm.fit`), and the density of a fit where you deliberately set $\mu = 0$ (wrong mean). Add a legend.

Exercise 2: Generate two samples: `sample_a` from $\mathcal{N}(5, 1)$ (size 30) and `sample_b` from $\mathcal{N}(5.5, 1)$ (size 30). Perform a two-sample t-test. Then repeat this experiment 1000 times and count how often the test rejects at level $\alpha = 0.05$. This is the empirical power of the test. How does it change if you increase the sample size to 100?

Exercise 3: Generate 200 samples from a mixture of two normals: with probability 0.4 draw from $\mathcal{N}(-2, 1)$, otherwise from $\mathcal{N}(3, 0.5^2)$. Write a function that computes the negative log-likelihood of a two-component Gaussian mixture (with 5 parameters: $\mu_1, \sigma_1, \mu_2, \sigma_2, p$). Use `scipy.optimize.minimize` to find the MLE. Plot the histogram and the fitted density.

Exercise 4: Load or generate a dataset with two numerical features and a binary target. Standardize the features. Implement logistic regression from scratch using gradient descent (see the archived Chapter 11 for guidance). Plot the decision boundary. Compare your accuracy with `scipy.optimize.minimize` applied to the negative log-likelihood.